

Balance of Activity during a Critical Period Tunes a Developing Network

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Abstract

Developing neural circuits are influenced by activity and are especially sensitive to changes in activity during critical periods (CPs) of development. Changes occurring during a CP often become 'locked-in' so that they affect the mature network. Indeed, several neurodevelopmental disorders have been linked to excessive activity during such periods. It is, therefore, important to identify those aspects of neural circuit development that are influenced by neural activity during a CP. In this study, we take advantage of the genetic tractability of *Drosophila* to show that activity perturbation during an embryonic CP permanently alters properties of the locomotor circuit. Specific changes we identify include increased synchronicity of motoneuron activity, and greater strengthening of excitatory over inhibitory synaptic drive to motoneurons. These changes are sufficient to reduce network robustness, evidenced by increased sensitivity to induced seizure. We also show that we can rescue these changes when increased activity is mitigated by inhibition provided by mechanosensory neurons. Similarly, we demonstrate a dose-dependent relationship between inhibition experienced during the CP, and the extent to which it is possible to rescue the hyperexcitable phenotype characteristic of the *para^{bss}* mutation. This suggests that developing circuits must be exposed to a properly balanced sum of excitation and inhibition during the CP to achieve normal mature network function. Our results, therefore, provide novel insight into how activity during a CP shapes specific elements of a circuit, and how activity during this period is integrated to tune neural circuits to the environment in which they will likely function.

eLife assessment

This **valuable** study combines electrophysiology and neuroanatomy with pharmacological and optogenetic manipulation in the *Drosophila* genetic model system to pinpoint the neural substrate that is influenced by altered activity during a critical period (CP) of larval locomotor circuit development. Increasing activity during the CP causes permanent network changes, manifesting in increased recovery times from seizures and altered intersegmental coordination during locomotion, thus indicating that a setpoint of network excitability is determined during the CP. Next, **compelling** experiments demonstrate that this goes along with increased excitation/inhibition ratios to single identified motoneurons and most importantly, for excitability setpoint determination during the CP excitatory and inhibitory inputs are integrated such that the effect of CP hyperexcitation is rescued by the stimulation of endogenous inhibitory inputs to the motoneurons. This provides novel insight into how developing neural network excitability is tuned and how it can be entrained during the CP.

Introduction

A developing nervous system is exposed to multiple, and unpredictable, influences that can alter mature circuit function. These include extrinsic and intrinsic factors, such as temperature, light, mechanical stress and/or genetic mutations. Multiple studies have revealed that developing neural networks are most sensitive to neuronal activity during critical periods (CPs) – defined periods in developmental time during which plasticity peaks (Hensch, 2005 [↗](#); Reh *et al.*, 2020 [↗](#); Wong-Riley, 2021 [↗](#)). Indeed, work in several model organisms has shown that activity perturbation during a CP often becomes ‘locked in’ so that once the CP is closed, the mature nervous system remains changed. This was first demonstrated by Hubel and Wiesel (Hubel and Wiesel, 1970 [↗](#)) who showed that blocking visual input in kittens, but not in adult cats, caused significant, lasting change in ocular dominance. Thus, visual activity during early life is necessary for normal development of vision. Since these pioneering experiments, CPs have been identified in other networks and across species, which indicates that they are a common phenomenon in the development of neural circuits (Marley and Baines, 2011a [↗](#); Giachello and Baines, 2015 [↗](#); Golovin *et al.*, 2019 [↗](#); Styr *et al.*, 2019 [↗](#); Hensch and Bilimoria, 2012 [↗](#)).

Recent studies posit that aberrant activity during CPs contributes to, and may even cause neurodevelopmental disorders including autism (LeBlanc and Fagiolini, 2011 [↗](#)), schizophrenia (Selemon and Zecevic, 2015 [↗](#)) and epilepsy (Lybrand *et al.*, 2021 [↗](#)). Specifically, this growing body of work suggests that excessive activity experienced during a CP may lead to increased excitability in a mature network. Research in several model organisms, including mice and *Drosophila*, supports this view by showing that mature neurons are more excitable when they experience greater activity during development (Marley and Baines, 2011a [↗](#); Giachello *et al.*, 2021 [↗](#); Styr *et al.*, 2019 [↗](#)). Despite this progress, however, the specifics of how activity experienced during a CP influences a nervous system remains unclear. Key questions that remain unanswered include: “how does activity perturbation affect the relative contribution of excitatory and inhibitory neurons within a network?”, and: “is it the extremes, or the sum of activity, experienced during a CP that is critical for mature circuit function?” It is difficult to answer these questions when conducting experiments in complex mammalian sensory networks, such as the visual and auditory networks, which have been extensively used to explore CPs (e.g., (Hooks and Chen, 2020 [↗](#); Nakamura *et al.*, 2020 [↗](#))). In this respect, the *Drosophila* larval network may provide a powerful alternative.

The larval nervous system is an experimentally-tractable and numerically simple model that is very well suited to studying CPs (Hunter *et al.*, 2021 [↗](#)). Larvae are amenable to a range of different assays to demonstrate the impact of manipulating activity during development on a mature nervous system. For example, the electroshock assay enables assessment of ‘network robustness’, by measuring the ability of the nervous system to withstand or recover from strong electrical stimulation (Marley and Baines, 2011a [↗](#)). Similarly, functional imaging can be used to visualise locomotor network dynamics (Streit *et al.*, 2016 [↗](#)). The *Drosophila* larval connectome project (which is near completion) has led to the mapping of individual neurons, neural circuits, and development of Gal4 lines (genetic driver lines) that enable cellular-level precision in experimental manipulation of the nervous system (reviewed in (Hunter *et al.*, 2021 [↗](#))). It is also possible to perform electrophysiology on certain cells, such as the anterior corner cell motoneuron (aCC, a.k.a MN-1b (Baines and Bate, 1998 [↗](#); Marley and Baines, 2011b [↗](#))). Together, these techniques and tools can be applied to study and attempt to answer questions including those posed above, by probing processes associated with an embryonic locomotor CP that was first described in previous work by our group (Giachello and Baines, 2015 [↗](#)).

Our earlier work showed that manipulating activity during 17-19h after egg laying (AEL) is sufficient to induce significant and permanent change to locomotor network stability. It also established that the range of manipulation that causes this change is broad; it extends from increased excitation during this CP being sufficient to induce a seizure-like phenotype in mature larvae subjected to electroshock, to a rescue of a genetic seizure mutation by reduction of network activity during the same period (Giachello *et al.*, 2015). However, the effects of activity perturbation on the specific cellular components of the locomotor network have not been identified. Here, we make use of the output of the connectome project to focus attention on excitatory cholinergic A18a (Hasegawa, Truman and Nose, 2016 [↗](#)) and inhibitory GABAergic A31k (Schneider-Mizell *et al.*, 2016 [↗](#)) interneurons, as well as the aCC motoneuron which these interneurons are monosynaptically connected to (Giachello *et al.*, 2022 [↗](#)). Specifically, we perform patch clamp experiments on aCC to show how synaptic connectivity of these single cell types (A18a or A31k) is influenced following activity perturbation during the embryonic CP. We show that excessive activity during the CP permanently increases the strength of the input of A18a relative to A31k, which is consistent with a significant change in the excitation: inhibition (E:I) balance of the network. Next, we show that restoring E:I balance rescues the network, thus providing evidence that the CP integrates the ‘averaged’ activity that occurs while it is open. Our data suggest that ‘average’ activity must be balanced, so that the intrinsic properties that dictate the function of the mature locomotor network are set within normal ranges.

Results

Increasing Neural Activity during Embryogenesis

Permanently alters Network Function and Synchronicity

To manipulate neural activity in developing embryos, we fed gravid (i.e. egg-laying) female flies picrotoxin (PTX), which increases neural activity by non-competitive antagonism of the GABA_A receptor (Lin *et al.*, 2012 [↗](#)). This procedure is sufficient to expose embryos and the resulting young larvae to PTX. Larval metabolism and excretion during early to mid-developmental stages removes PTX, so that it is no longer detectable when assaying for effect in wall-climbing third instar larvae (L3, (Marley and Baines, 2011a [↗](#))). Crucially, our previous work has shown that permanent changes in network function observed as a result of embryonic exposure to PTX occur due to the change in activity during the CP, which lasts from 17-19h AEL (Giachello and Baines, 2015 [↗](#)). Similar activity-manipulation during larval stages is without effect (Sarah Doran and RAB, unpublished data). We measured the effect of PTX-linked activity manipulation during the CP on two independent parameters at L3: recovery from electroshock (which is longer in other models of CP activity manipulation (Giachello and Baines, 2015 [↗](#))) and crawling velocity (**Figure**

1A-C). Videos of seizure activity can be viewed in Marley and Baines, (2011). Embryonic exposure to PTX significantly increased the duration of electroshock-induced seizures compared to controls (61.11 ± 3.94 s, EtOH; 96.49 ± 6.81 s, PTX; $n = 36, 35$; $p < 0.0001$) (**Figure 1B**). PTX exposure did not, however, influence crawling velocity (1.06 ± 0.04 mm/s, EtOH; 0.98 ± 0.04 mm/s, PTX; $n = 56, 58$; $p = 0.14$) (**Figure 1C**). This suggests that changes caused by PTX exposure during the CP were not significant enough to alter circuit function under ‘normal’ conditions (i.e. crawling) but clearly altered network robustness: the ability of the network to respond to strong stimulation (electroshock).

With a network-level phenotype for perturbation of CP activity established (increased seizure), we tested for possible correlates at the level of circuit activity within the locomotor network. In larvae, normal locomotor function is characterised by stereotyped activity that is regulated by a central pattern generator (CPG (Pulver *et al.*, 2015)). This pattern is formed by activity-waves that pass between successive segments in the ventral nerve cord (VNC) to support peristaltic locomotion. The timing (or synchronicity) of waves crossing adjacent segments is characteristic and can be measured as the time lag between peaks of activity observed in adjacent aCC motoneurons. Previous work has shown that synchronicity increases (i.e., time lag reduces) in the presence of mutations that predispose animals to seizure (e.g. *para^{bss}*), or following acute exposure to proconvulsants (e.g. 4-aminopyridine or PTX (Streit *et al.*, 2016)). We therefore quantified synchronicity of activity between aCC motoneurons at L3, to investigate how embryonic exposure to PTX alters activity in the mature locomotor network (**Figure 1D-F**).

Increasing neural activity during embryogenesis (PTX) significantly increased synchronicity across segments A6-A4, compared to vehicle control (0.9 ± 0.09 s, EtOH; 0.67 ± 0.06 s, PTX; $n = 18, 18$; $p = 0.05$) (**Figure 1F**). This increased synchronicity of activity between adjacent segments is consistent with, and may contribute to, the seizure phenotype observed. It likely reflects significant changes in CPG properties that make it harder for larvae to recover from electroshock.

Increased Activity during the CP Alters the E:I Balance within the Locomotor Network

Motoneuron activity reflects the integration of the excitatory and inhibitory premotor synaptic drive to which the cell is exposed. We therefore tested the effect of embryonic exposure to PTX on the synaptic drive provided by two identified premotor interneurons – one exciter and one inhibitor - which are monosynaptically connected to aCC (Giachello *et al.*, 2022). Specifically, we quantified the strength of cholinergic excitor A18a (a.k.a. CLI2, (Hasegawa, Truman and Nose, 2016)) and GABAergic inhibitor A31k (Schneider-Mizell *et al.*, 2016) to aCC at L3, following PTX-induced activity perturbation during embryonic development. To achieve this, premotor interneurons were activated using optogenetics, while whole-cell patch recordings were made from aCC (**Figure 2A**, left-hand panel).

Larval motoneurons do not generate action potentials (APs) without synaptic drive. Thus, to determine how synaptic inputs may have changed following embryonic CP manipulation, current was injected into aCC to evoke stable firing before premotors were stimulated. Evoked firing before optogenetic stimulation, in larvae expressing A18a-Gal4, was different between control L3 (EtOH) and those exposed to PTX (6.87 ± 0.46 Hz, EtOH; 10.41 ± 1.44 Hz, PTX; $n = 8, 8$; $p = 0.03$). While this difference may reflect a change in excitability due to PTX exposure, it may also have been due to the requirement for injecting differing amounts of current to maintain spiking at a constant rate. Regardless, optogenetic activation of excitatory A18a increased AP frequency in aCC in control L3 (**Figure 2A**, right-hand panel) and this increase was significantly larger in L3 derived from embryos exposed to PTX (7.24 ± 2.04 Hz, EtOH; 15.81 ± 3.14 Hz, PTX; $n = 8, 8$; $p = 0.04$) (**Figure 2B**). Evoked firing before optogenetic stimulation, in larvae expressing A31k-Gal4, was not significantly different between control L3 (EtOH) and those exposed to PTX (7 ± 0.94 Hz, EtOH; 7.36 ± 1 Hz, PTX; $n = 8, 9$; $p = 0.8$). Activation of inhibitory A31k decreased AP firing in control L3 (**Figure**

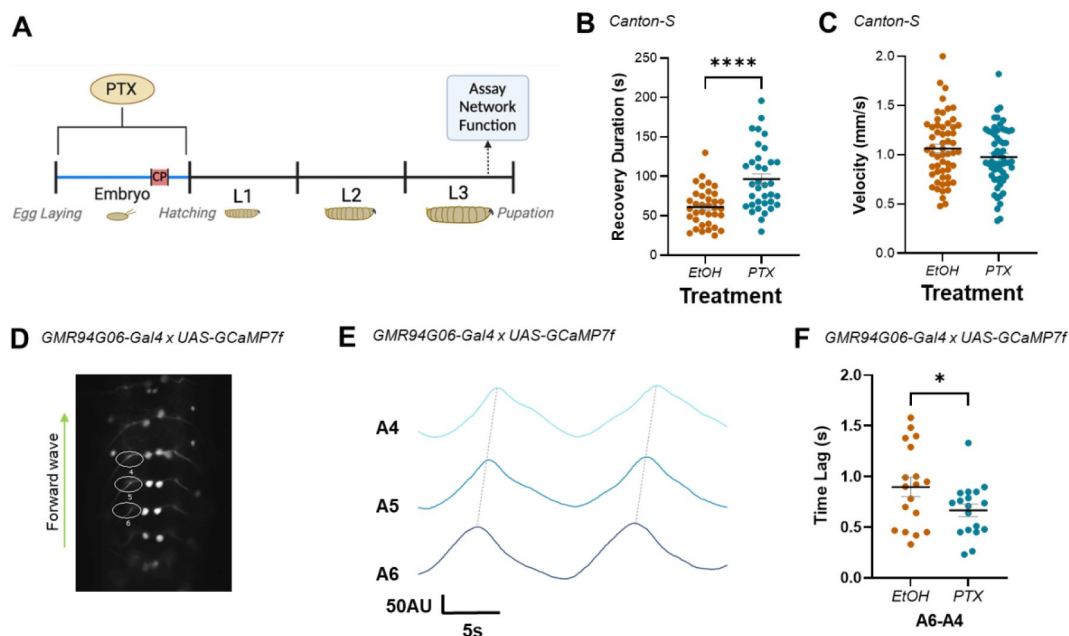


Figure 1

Increased excitation during the embryonic CP alters network stability and motoneuron network synchrony

(A) Schematic demonstrating timeline for picrotoxin (PTX) exposure, and assay of network function. PTX-induced increase in activity during development increases third-instar larval duration of recovery from induced seizure (B) but does not change crawling velocity (C). (D) ROIs captured GCaMP signal generated by forward waves passing through ipsilateral aCC motoneuron axons in the L3 ventral nerve cord, abdominal segments (A6-4). (E) Representative traces recorded in 3 adjacent abdominal segments (A6-A4), during 2 forward waves. Colours represent different segments. Synchrony was measured as the time lag (s) of activity passing between adjacent segments (i.e., between peaks, dotted lines). (F) Embryonic exposure to PTX caused a significant increase in synchronicity across segments A6-A4 versus EtOH controls.

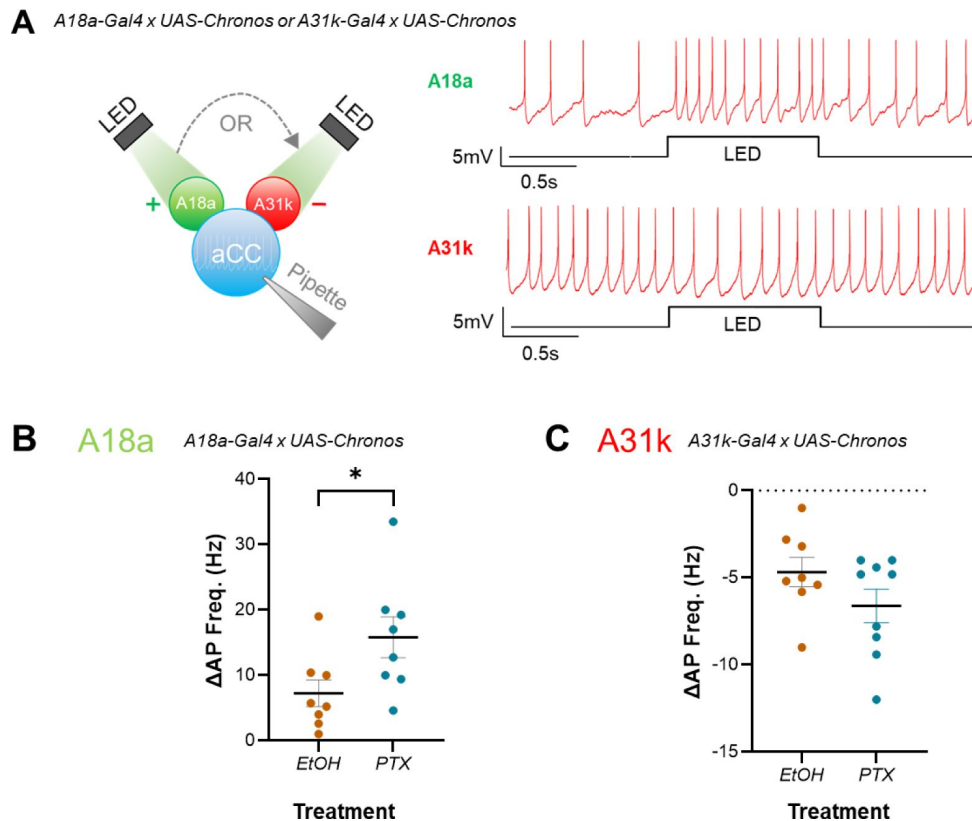


Figure 2

Increased excitation during the CP preferentially increases the strength of excitatory inputs to aCC

(A) Left-hand panel: schematic demonstrating use of optogenetics for testing strength of A18a or A31k synaptic drive on evoked action potential firing in aCC. Right-hand panel: representative traces illustrating the effect of excitatory or inhibitory input on evoked action potential firing in aCC, respectively. (B, C) current-clamp recordings wherein current (~10pA) was injected to evoke action potential firing in aCC, prior to optogenetic stimulation of premotor input. Amplitude of inputs was quantified as change in frequency of action potential firing (Δ AP Freq. (Hz)) following optogenetic stimulation of A18a or A31k. Change in action potential firing frequency in A18a ($p = 0.04$), but not A31k ($p = 0.15$), was significantly potentiated by embryonic exposure to PTX vs vehicle controls (EtOH).

2A), however, this decrease was not significantly different in L3 derived from embryos exposed to PTX ($23.71 \pm 6.76\text{Hz}$, EtOH; $28.24 \pm 2.4\text{ Hz}$, PTX; $n = 8, 9$; $p = 0.15$ (**Figure 2C**). Comparing the magnitude of the changes in A18a or A31k input using an effect size calculation (see Methods for details) corroborates that excitation increased more than inhibition, as Cohen's $d = 1.14$ (large effect size) for A18a and 0.78 (medium effect size) for A31k. Measurement of V_m , R_{in} and capacitance showed no significant changes due to the presence of PTX (V_m : $-47.8 \pm 1.26\text{mV}$, EtOH; $-47.6 \pm 0.78\text{Hz}$, PTX; $n = 16, 17$; $p = 0.84$); (R_{in} : $765.4 \pm 27.05\text{M}\Omega$, EtOH; $746.1 \pm 28.13\text{ M}\Omega$, PTX; $n = 16, 17$; $p = 0.62$); (cap: $23.2 \pm 1.0\text{pF}$, EtOH; $23.1 \pm 2.6\text{pF}$, PTX; $n = 16, 17$; $p = 0.95$). Our data suggests that embryonic exposure to PTX induced a shift in the E:I balance of the larval locomotor circuit to favor excitation, which is consistent with the network-level phenotypes observed (**Figure 1**). Thus, our results suggest that increasing activity during the embryonic CP, increases the strength of excitatory, but not inhibitory, premotor synaptic drive.

Increased Excitatory Input to Motoneurons Reflects Increased Postsynaptic Sensitivity to Quantal Release

Increased excitatory synaptic drive to aCC may be due to either, or both, presynaptic and postsynaptic adaptations to manipulation of activity during embryogenesis. Thus, we sought to understand which of these effects may be responsible for the changes we observed. A change in the frequency of mini excitatory postsynaptic potentials (mEPSPs, a.k.a. minis) would suggest the adaptation is primarily presynaptic (e.g. increased probability of release), whilst a change in distribution and/or amplitude of minis is more consistent with a mechanism acting postsynaptically (e.g. increased or altered receptor subunits). We quantified the mean amplitude, cumulative distribution, and frequency of minis. Specifically, we began by recording spontaneous minis in wild-type larvae following embryonic exposure to PTX (**Figure 3A**).

Whole-cell voltage-clamp recordings, made in $2\mu\text{M}$ tetrodotoxin (TTX, which limits synaptic release to minis) and with V_m held at -60mV , showed that manipulating activity during embryogenesis with PTX, did not statistically change mini frequency ($5.01 \pm 1.12\text{Hz}$, EtOH; $2.96 \pm 0.57\text{Hz}$, PTX; $n = 7, 8$; $p = 0.11$) (**Figure 3B**). In contrast, PTX treatment led to a significant increase in mini amplitude ($4.08 \pm 1.42\text{pA}$, EtOH; $4.51 \pm 1.71\text{pA}$, PTX; $n = 188, 199$; $p = 0.008$) (**Figure 3C**), causing a corresponding rightward shift in the cumulative frequency distribution (%) of currents recorded (**Figure 3D**). Thus, our recordings of minis in aCC are consistent with a change in the postsynaptic response to excitatory quantal release. However, this conclusion is based on measuring all excitatory quantal inputs to aCC, and not specifically from A18a. It was, therefore, important to refine this result by restricting it to the specific excitatory quantal input from this premotor interneuron.

It is technically challenging to record minis elicited in response to quantal release from a defined neuron, because doing so requires isolating minis from a single neuron (e.g. A18a) from those released by other excitatory premotors (there are many which synapse with aCC (Zarin *et al.*, 2019)). To maximize our ability to do so, we adapted a technique developed in mammals, which substitutes extracellular Sr^{2+} (8mM) for Ca^{2+} (2mM) (Xu-Friedman and Regehr, 1999; Bekkers and Clements, 1999). This substitution promotes asynchronous vesicular release (i.e., occurrence of single quantal events) following firing of an evoked action potential by a presynaptic neuron, which can be recorded in a postsynaptic neuron as a series of minis. Thus, we replaced Ca^{2+} (2mM) with Sr^{2+} (6mM , which better supported survival of larval neurons than 8mM) and used optogenetics to stimulate A18a, while recording from aCC. We validated this approach by showing that optogenetic activation of A18a, in the presence of Sr^{2+} , greatly reduced evoked synaptic current but increased the frequency of minis recorded in aCC (**Figure S1**), which is consistent with asynchronous release. Thus, under these conditions, stimulation of A18a produced a series of minis predominantly released from this premotor interneuron (**Figure 4A**).

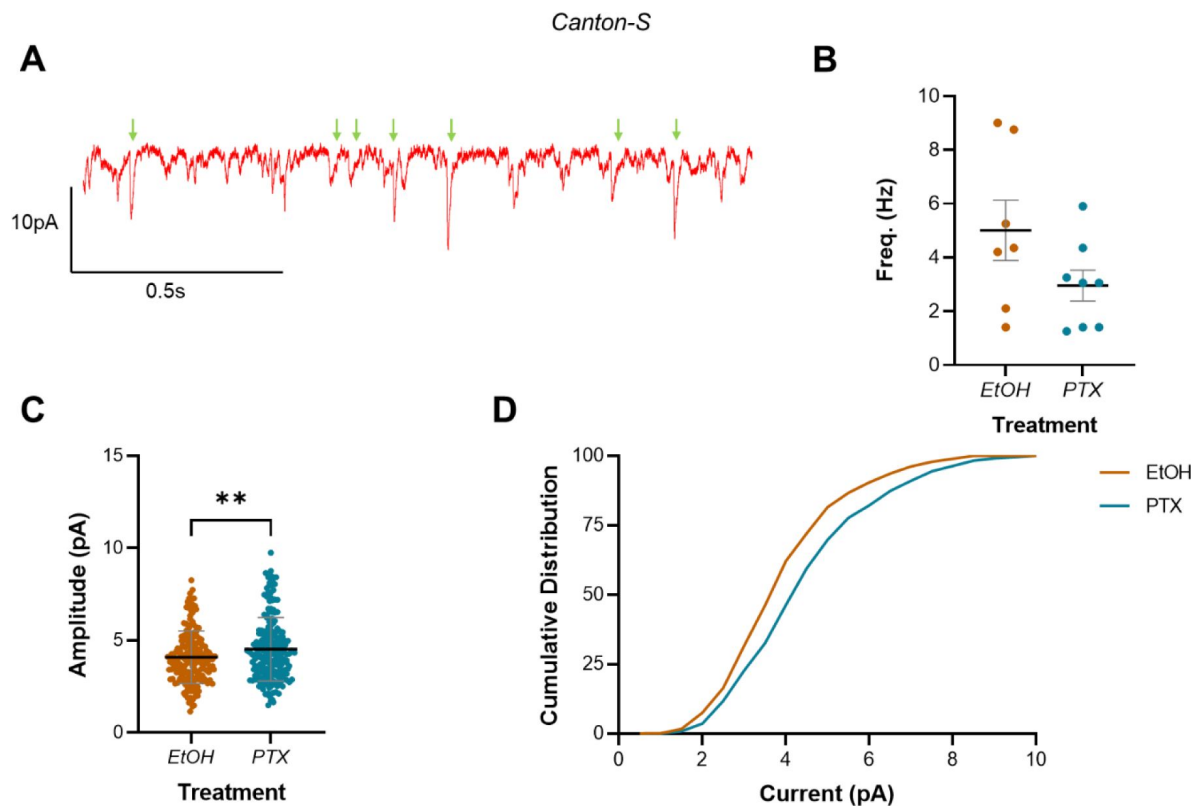


Figure 3

Increasing activity during the embryonic CP increased mini amplitude but not frequency in L3 aCC

(A) Representative trace showing minis (green arrows) in L3 aCC. Events not marked are not mini's and probably represent single channel events (see Fig S1D for an expanded version of this same trace). (B) Mini frequency, as events observed in the first 20s of a whole-cell voltage-clamp recording (V_m held at -60mV), show no difference between vehicle controls (EtOH) and L3 exposed to picrotoxin (PTX) during embryogenesis. (C) Mini amplitude from the same recordings reveal significantly larger events in larvae that were exposed to PTX during embryogenesis, vs. vehicle controls (EtOH). (D) Cumulative distribution (%) of mini amplitude demonstrates a rightward shift in current size, following activity manipulation.

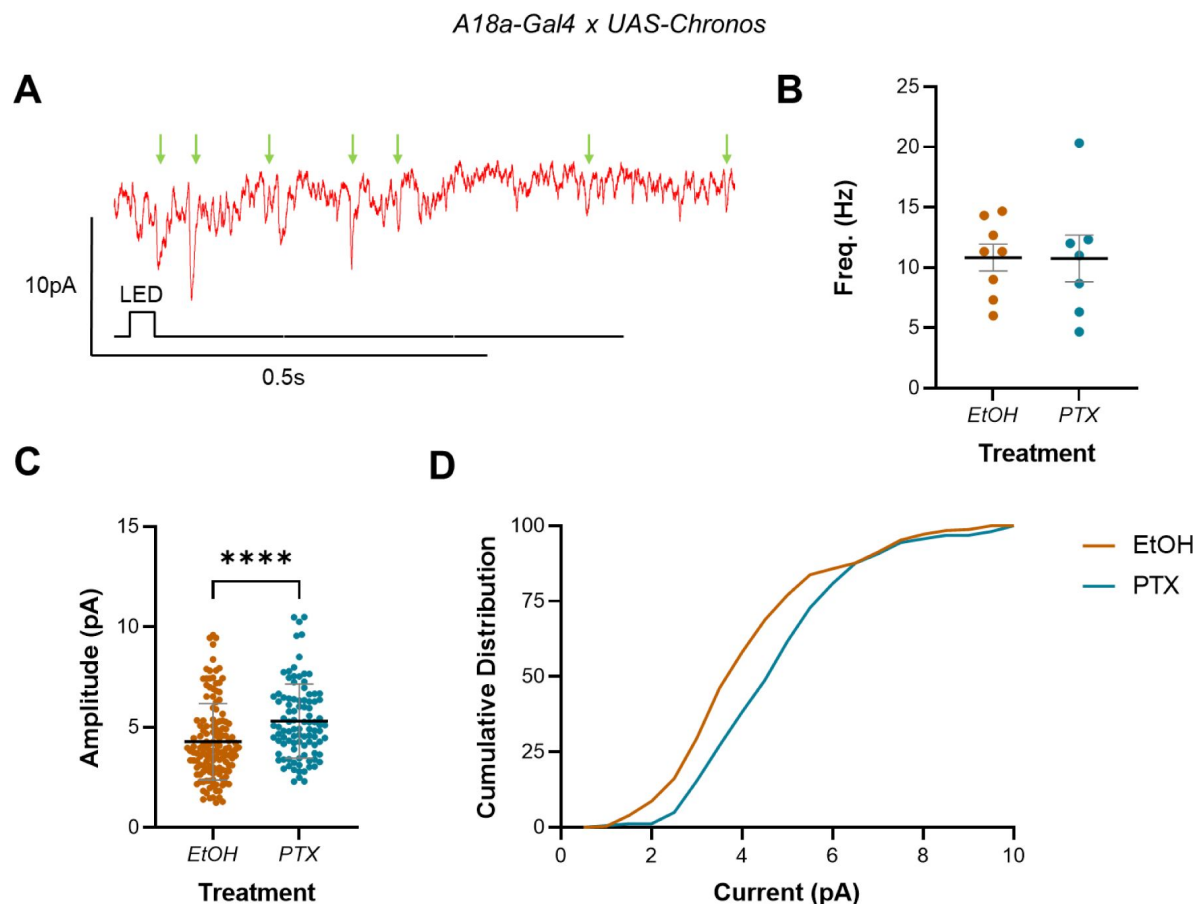


Figure 4

Increasing activity during the embryonic CP increased A18a-specific mini amplitude but not frequency in L3 aCC

(A) Representative traces showing minis (mEPSPs, green arrows) in L3 aCC (V_m held at -60mV), following optogenetic excitation of A18a. (B) Mini frequency, measured as events observed in the 3s following optogenetic stimulation of A18a (50ms light pulse) was not different between vehicle-only controls (EtOH) and L3 exposed to picrotoxin (PTX) during development. (C) Mini amplitude was significantly larger following exposure to PTX during embryogenesis, versus controls (EtOH). (D) Cumulative distribution (%) of mini amplitudes demonstrates a clear rightward shift in current size, following activity manipulation.

Analysis of whole-cell patch-clamp recordings (V_m held at -60mV) showed that there was no change in mini frequency, in the 3s following optogenetic stimulation of A18a, in PTX versus controls ($10.83 \pm 1.12\text{Hz}$, EtOH; $10.76 \pm 1.93\text{Hz}$, PTX; $n = 8, 7$; $p = 0.97$) (**Figure 4B** [↗](#)). There was, however, a significant increase in mini amplitude ($4.28 \pm 1.91\text{pA}$, EtOH; $5.31 \pm 1.85\text{pA}$, PTX; $n = 138, 98$; $p < 0.0001$) (**Figure 4C** [↗](#)), and corresponding rightward shift in the cumulative distribution of current amplitudes (**Figure 4D** [↗](#)). These data are consistent with a change in the postsynaptic sensitivity of aCC to quantal release, which alters the E:I balance in the L3 locomotor circuit, following manipulation of activity during embryogenesis. Moreover, that we see the same effect in the presence of external Ca^{2+} (**Figure 3** [↗](#)) likely indicates that sensitivity to all excitatory inputs has increased.

Increased Excitation during the CP does not Alter Motoneuron Morphology or Synaptic Connectivity

Given our mini recordings suggested that the change in E:I balance may be associated with a change in the postsynaptic response to neurotransmitter release, we used high-resolution confocal microscopy to characterise the morphology and synaptic connectivity of aCC, following embryonic treatment with PTX (**Figure 5** [↗](#)). This analysis demonstrated no obvious change in dendritic arbor length, number of branch points or terminals (**Figure 5E** [↗](#)). Furthermore, the number of excitatory postsynaptic sites (Drep2 puncta number) and synaptic density (μM dendrite per synapse) of aCC was unchanged following exposure to PTX during embryogenesis (**Figure 5F** [↗](#)). This suggests that the changes in mini amplitude, other physiology, and seizure phenotype we observed, are not due to significant change in cellular morphology. Rather, it suggests the changes may be related to an alteration in excitatory postsynaptic receptor number and/or sub-unit composition.

Stimulating Mechanosensory Neurons during the CP Rescues the Seizure Phenotype Associated with Excessive Excitatory Activity

As described above, developing circuits can be exposed to various influences (light, temperature, mutations etc.) which may alter levels of activity. While some of these influences will increase excitation, others may decrease it. Because our data suggest that experiencing excessive excitation during the CP leads to an imbalanced mature network, it is conceivable that the nervous system must experience balance or a particular sum of excitatory and inhibitory influences, during the CP, to encode normal network properties ([Fagiolini and Hensch, 2000](#) [↗](#)).

To test this hypothesis, we combined manipulations of predicted opposite valence to see whether, in the instance that we could achieve approximately equal strengths, increased excitation and inhibition would cancel each other. We chose to stimulate chordotonal (ch) neurons to counteract PTX. This is because ch neurons are mechanosensitive cells that have been proposed to provide an inhibitory ‘brake’ that slows locomotor network activity during the CP ([Carreira-Rosario et al., 2021](#) [↗](#)). We, therefore, built a device to stimulate ch neurons (see Methods and Figure S2) and performed experiments in which they were activated during the CP (17-19hrs AEL) or in immediately adjacent periods (15-17h or 19-21h AEL) to assess the impact of the brake alone, before pairing it with PTX (**Figure 6A-B** [↗](#)).

Activation of ch neurons during the CP, but not before or after it, significantly lengthened recovery from induced seizures in L3 larvae ($72.69 \pm 6.02\text{s}$, 15-17h AEL; $122.7 \pm 8.43\text{s}$, 17-19h AEL; $65.83 \pm 26.07\text{s}$, 19-21h AEL; $n = 39, 38, 40$; $p < 0.0001$ for 15-17h vs. 17-19h and 17-19h vs. 19-21h, 0.58 for 15-17h vs. 19-21h) (**Figure 6B** [↗](#)). This result shows that the majority of ch neuron activity influence on mature circuit function is, like PTX, limited to the embryonic CP. Similarly, that stimulating ch activity during the CP increases the seizure response to electroshock, is consistent with optogenetic inhibition causing the same effect ([Giachello and Baines, 2015](#) [↗](#)). This result also

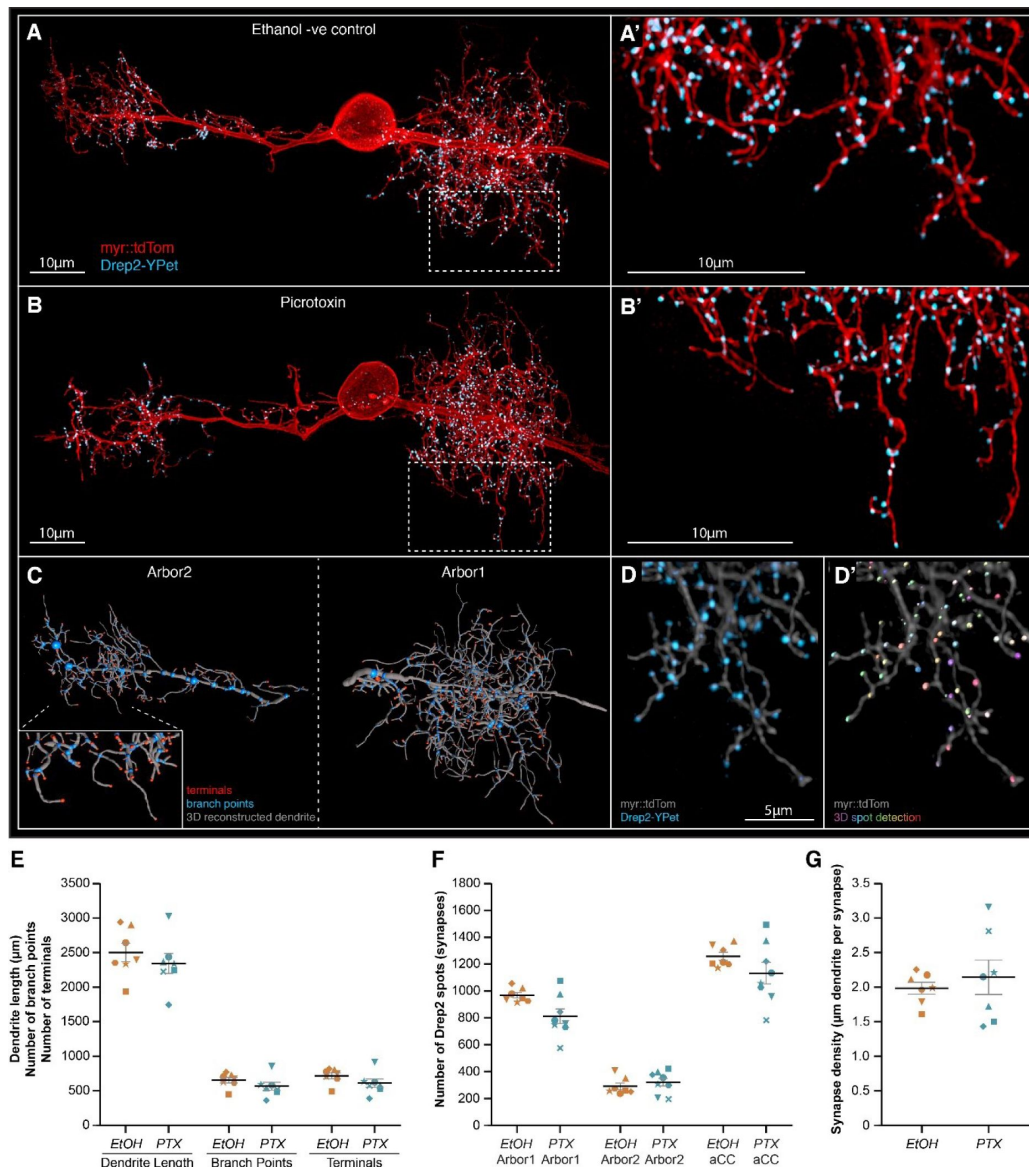


Figure 5

Increasing neural network activity by PTX exposure during development does not alter motoneuron dendritic arbor size or excitatory postsynaptic sites

(A-B) Representative images of aCC motoneurons from larval ventral nerve cords 72 hours after larval hatching (ALH), from control (A) or embryonic exposure to picrotoxin (B). aCC dendritic arbor structure is marked by membrane targeted myr::tdTomato (red), and postsynaptic excitatory sites are marked by a Drep2::YPet fusion protein (cyan). Visualisation of individual aCC motoneurons is achieved by an RN2-enhancer element driving a stochastic FLP-out LexA expression system. (C) tdTomato fluorescence was used for AI-supported 3D reconstruction of the whole aCC dendritic arbor, allowing quantification of total dendrite length (grey segments), branch points (blue spots) and terminals (red spots). (D) Following live imaging, nerve cords were fixed and stained using a fluorophore-tagged nanobody to detect the Drep2::YPet fusion protein (D), enabling high resolution and photostable imaging of Drep2::YPet puncta for quantification of excitatory postsynaptic site number by semi-automated spot detection (D'). (E) The total dendrite length, number of branch points, and number of terminals are shown, for both control and picrotoxin treatments. (F) The number of Drep2::YPet puncta for the larger ipsilateral arbor ('arbor1') the smaller contralateral arbor ('arbor2') and the combined sum across both arbors ('aCC') is shown, for both control and picrotoxin treatments. (G) Synapse density, calculated as the total dendrite length divided by total Drep2::YPet positive postsynaptic sites. In E-G, each point represents a single aCC cell, where the symbol shape denotes the particular cell.

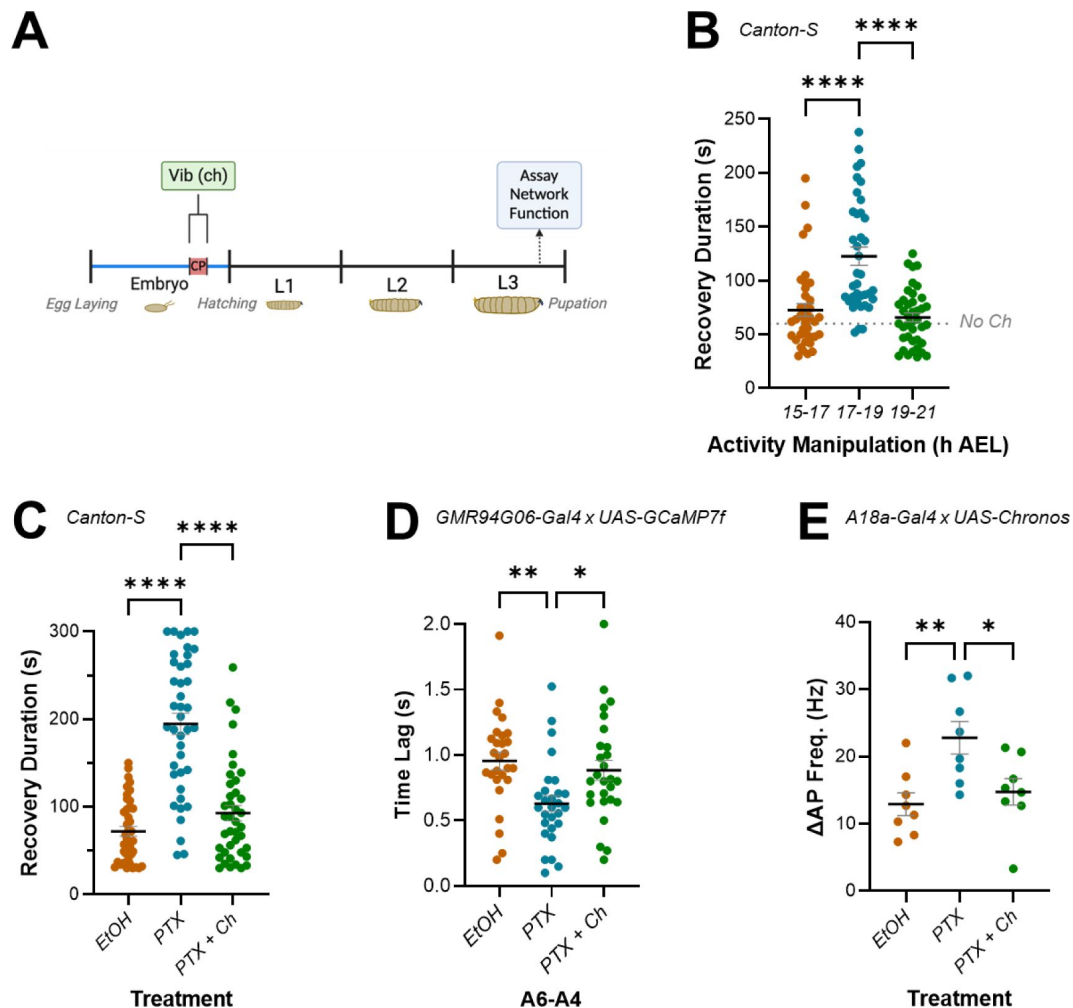


Figure 6

Activating mechanosensory neurons during the CP rescues E:I balance due to increasing activity during embryogenesis

(A) Schematic showing stimulation of mechanosensory chordotonal neurons (ch) by vibration during the CP (17-19h after egg laying, AEL). (B) Stimulating ch neurons during the CP, but not during 15-17h AEL or 19-21h AEL, significantly increases mature larval (L3) recovery duration from electroshock-induced seizure. CS is unmanipulated wild-type (Canton-S) controls. (C-E) Exposure to picrotoxin (PTX) during embryogenesis significantly increases mature larval (L3) recovery duration from electroshock-induced seizure, increases network synchronicity, and increases the strength of excitatory (A18a) inputs to motor neurons (aCC). All these effects are rescued by co-activation of ch neurons during the CP (PTX + Ch).

supports previous work that showed perturbing activity of sensory neurons during the locomotor CP causes lasting changes to behaviour (Fushiki, Kohsaka and Nose, 2013; Carreira-Rosario *et al.*, 2021).

Next, we tested how ch activity impacted the phenotypes observed for recovery from electroshock, network synchronicity, and amplitude of inputs from A18a to aCC, following embryonic exposure to PTX (**Figure 6C-E**). Remarkably, ch neuron activation provided a complete rescue of both network and cellular-level phenotypes associated with embryonic exposure to PTX. Thus, recovery from electroshock, which was again increased following exposure to PTX, was rescued to control levels when PTX was paired with ch neuron activation (72.03 ± 5.71 s, EtOH; 194.7 ± 12.22 s, PTX; 92.75 ± 8.95 s, PTX + Ch; $n = 40, 40, 40$; $p < 0.0001$ for EtOH vs. PTX and PTX vs. PTX + Ch, 0.58 for EtOH vs. PTX + Ch) (**Figure 6C**). PTX-induced increase in synchrony of activity between adjacent aCC motoneurons was also rescued (0.96 ± 0.07 s, EtOH; 0.63 ± 0.06 s, PTX; 0.88 ± 0.08 s, PTX + Ch; $n = 27, 28, 27$; $p = 0.004$ for EtOH vs PTX, 0.75 for EtOH vs. PTX + Ch, 0.03 for PTX vs. PTX + Ch) (**Figure 6D**). Finally, the increased strength of excitatory A18a inputs to aCC following exposure to PTX, was rescued by stimulation of ch neurons during the CP (12.92 ± 1.704 Hz, EtOH; 22.79 ± 2.42 Hz, PTX; 14.75 ± 1.98 s, PTX + Ch; $n = 8, 8, 8$; $p = 0.008$ for EtOH vs PTX, 0.53 for EtOH vs. PTX + Ch, 0.023 for PTX vs. PTX + Ch) (**Figure 6E**). These results support the hypothesis that CPs integrate the excitatory and inhibitory activity to which they are exposed, to compute a sum that leads to a corresponding output in the mature locomotor network. If that sum equals balance (i.e., appropriate relative amounts of each), normal physiological function is established. An imbalance, however, may lead to altered network function.

Balancing Activity during the CP Rescues Adult Behaviour caused by Genetic Mutation

Next, we sought further confirmation that CPs integrate the activity they are exposed to. Specifically, we tested the effect of different amplitudes of ch neuron activity (as a function of 0, 40dB, 60dB and 80dB speaker volume), on recovery from seizure in bang sensitive (*para^{bss}*) L3. *para^{bss}* mutants express a hypermorphic variant of the sole *Drosophila* voltage-gated sodium channel (*para*), which makes neurons more excitable. Previous work has shown that excitation is increased at all stages of embryonic and larval development, resulting in increased synchrony between aCC neurons and increased duration of electroshock-induced seizures (Marley and Baines, 2011a; Streit *et al.*, 2016; Giachello and Baines, 2015). These larvae, therefore, provided an alternative means of inducing excessive excitation during the embryonic CP (rather than PTX) that we could oppose by ch neuron activity, to test for impact on recovery from electroshock (**Figure 7**).

In agreement with previous work (Giachello and Baines, 2015), *para^{bss}* recovery from electroshock was significantly longer than wild-type controls (168 ± 6.96 s, *bss*; 87.74 ± 3.45 s, CS; $n = 50, 50$; $p < 0.0001$). This difference was rescued completely when ch neurons were stimulated during the CP at 80dB (87.74 ± 3.45 s, CS; 94.51 ± 4.59 s, *bss* + 80dB; $n = 50, 50$; $p = 0.91$). This was the maximum strength of stimulation possible, and importantly, the effect was reduced in a dose-dependent manner as the intensity of ch stimulation during the CP decreased (125.6 ± 5.82 s, *bss* + 60dB; 144.3 ± 6.61 s, *bss* + 40dB; $n = 50, 50$; $p = 0.03$ for *bss* vs *bss* + 40dB, < 0.0001 for *bss* vs. *bss* + 60dB) (**Figure 7A**).

A prediction of the hypothesis that normal network function depends on a proper E:I balance during the CP is that reduced excitation during the CP would require proportionately reduced inhibition to counteract it. To test this, we raised *para^{bss}* at 21°C, which reduces the duration of evoked seizure when compared to animals raised at 25°C (168 ± 6.96 s, *bss* 25; 113.2 ± 4.28 s, *bss* 21; $n = 40, 50$; $p < 0.0001$). This is presumably due to reduced levels of excitation at this developmental temperature. At 21°C, stimulating ch activity during the CP using 40dB is sufficient to completely suppress seizure behaviour (76.24 ± 5.07 s, *bss* 21 + 40dB; 86.68 ± 4.2 s, CS; $n = 40, 40$; $p < 0.44$).

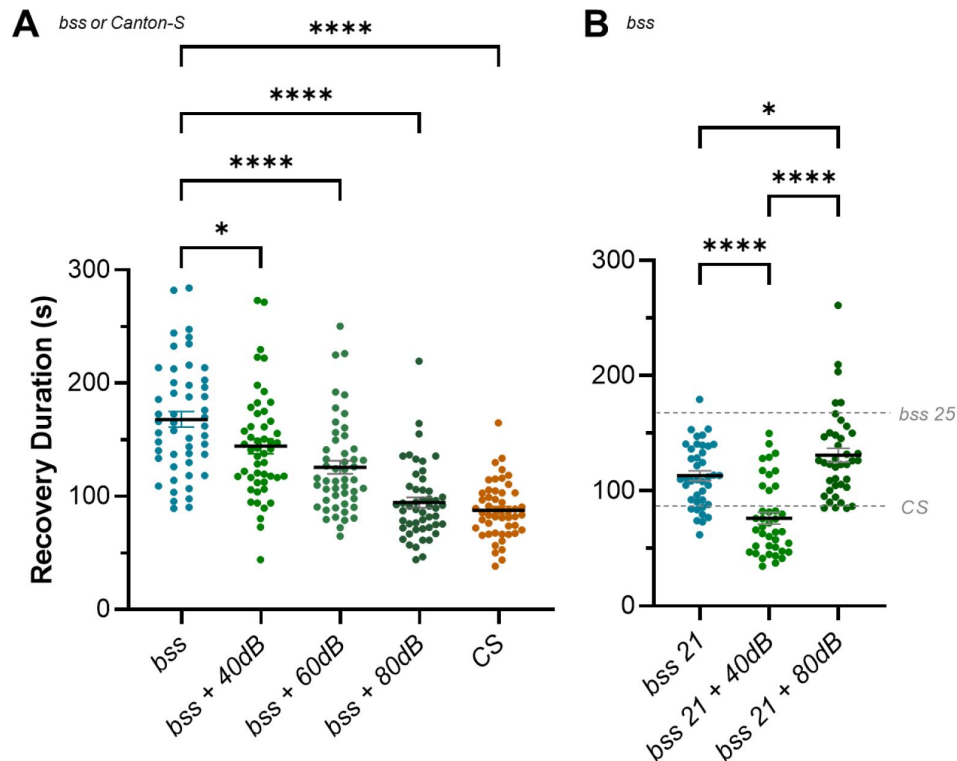


Figure 7

Restoring E:I balance during the CP rescues seizure phenotype caused by a genetic mutation

(A) The *para^{bss}* (*bss*) mutation causes lifelong neural hyperactivity, which is evident in a long duration of recovery from seizure when compared to wild-type controls (*Canton-S*). This recovery can be rescued by balancing the hyperactivity present with inhibition provided by sufficiently strong stimulation of ch neurons, during the CP (*bss* + 80dB). Weaker stimulation of the ch neuron-mediated inhibition provides intermediate phenotypes (*bss* + 60dB, *bss* + 40dB) that suggest the function of the mature network reflects the sum of excitatory and inhibitory activity during the CP. (B) *para^{bss}* raised at 21°C experience less excitation than those raised at 25°C (e.g. in (A)) during development, and demonstrate a reduced seizure phenotype compared to the latter. This phenotype is completely rescued by 40dB stimulation of ch neurons (vs. 80dB at 25°C (A)), which supports the idea that it is a balanced sum of excitatory and inhibitory activity during the CP, that enables normal mature network function. Similarly, exposing *para^{bss}* raised at 21°C to 80dB vibration during the CP results in a significant increase in seizure duration. We hypothesize that at this temperature, inhibition provided by ch neuron activity is greater than the increase in excitation resulting from the *para^{bss}* mutation.

Notably, ch activity at 80dB induces a seizure phenotype ($130.8 \pm 5.6s$, *bss* 21 + 80dB; $86.68 \pm 4.2s$, CS; $n = 40$, 40; $p < 0.0001$), which is consistent with 80dB ch-induced inhibition outweighing *para*^{bss}-induced excitation at 21°C (**Figure 7B**). Combined, these results support the hypothesis that the CP integrates the excitatory and inhibitory activity to which the locomotor network is exposed, using the sum (relative balance, or E:I) to set key network parameters that are then ‘locked in’ to the mature locomotor network. Proper balance leads to normal function, while imbalance may cause dysfunction.

Discussion

A large body of work, mostly performed in rodents, has shown that maturation of neural circuits is dependent on an interplay between excitatory and inhibitory activity, particularly during developmental CPs (Wong-Riley, 2021; Hensch, 2005; Hensch and Bilimoria, 2012; Gervain *et al.*, 2013). CPs often reflect a change from spontaneous to patterned activity within neural networks (Coulson *et al.*, 2022). Early spontaneous activity that occurs prior to CP opening is predominantly excitatory and notably includes excitatory GABAergic signalling (Ben-Ari, 2002). Thus, the E:I balance is strongly biased towards excitation prior to the opening of a CP (Allene *et al.*, 2008). This early developmental phase seemingly facilitates the development of neuronal electrical properties and synaptic connectivity (Baines and Pym, 2006). However, as sensory input becomes active, inhibitory signalling increases (with GABA now assuming its mature inhibitory signalling mode). This coincides with, and may enable, CP opening (Hensch, 2005; Hensch and Bilimoria, 2012).

Key outstanding questions related to CPs are: which network properties are set whilst a CP is open? And are these properties, which are often ‘locked in’, deterministic for the trajectory of a developing network? Our work provides further evidence that appropriate E:I during a CP is necessary for normal function in mature neural networks, which supports previous work (Giachello and Baines, 2015). Importantly, however, our work also presents novel insight into the precise role that appropriate E:I balance contributes during CPs at a level of synaptic detail that is difficult to attain in other models. We achieved this insight by exploiting the genetic tractability (courtesy of tools developed as a consequence of the *Drosophila* connectome project (Ohyama *et al.*, 2015; Zheng *et al.*, 2018; Scheffer *et al.*, 2020)) and accessibility of the larval nervous system (reviewed in Hunter *et al.*, 2021), to demonstrate that increasing activity during the CP alters mature network E:I to affect permanent change to locomotor network function.

Collectively, our results show that activity manipulation during a CP alters the E:I balance in defined components of the larval locomotor circuit. Specifically, we observe that the strength of input from an excitor (cholinergic A18a) to a motoneuron (aCC) increased more than the input from an inhibitor (GABAergic A31k). However, we recognise that other elements of the locomotor network may also undergo change due to CP manipulation. The advantage of this system is that most of these elements can be manipulated through cell-specific genetic drivers. The firing frequencies that we imposed (approx. 5 – 10 Hz) are also lower than seen during fictive locomotion (Kadas *et al.*, 2017), which shows burst firing lasting for ~300 ms and achieving spike frequencies of up to 100Hz. Nevertheless, we identified an increase in the amplitude of minis which suggests the primary mechanism that alters excitability is postsynaptic. The precise mechanism contributing to increased mini amplitude remains to be determined, but a plausible scenario may involve change in cholinergic subunit composition. Similar changes to synaptic physiology have been reported in mammalian systems. For example, monocular deprivation in rat increases excitability in layer 4 of the visual cortex skewing the E:I balance towards excitation (Maffei, Nelson and Turrigiano, 2004). However, the complexity of the visual system obscures definitive conclusions because of apparent confounding observations of increasing inhibition in different components of the same circuitry (Maffei, Nelson and Turrigiano, 2004). Interestingly, these observations mirror our results where we report increases in both excitation and inhibition, but

show that the former predominates, thus altering E:I balance. Moreover, and again similar to our findings, monocular deprivation in rats is sufficient to increase mini amplitude indicative of active postsynaptic mechanisms for altered excitability. Specifically, mini amplitude first declines following monocular deprivation, but after 3-6 days is potentiated due to homeostatic synaptic scaling (Lambo and Turrigiano, 2013 [↗](#)). This observation suggests that homeostatic compensation can alter properties set during a CP and thus the final effect to any given network is likely to be due to a mix of CP-dependent and homeostatic-dependent change. The ability to record from identified network components and capability to explore along the entire developmental timeline of the *Drosophila* larval locomotor circuit offers the prospect of exploring this in greater detail. For example, it remains unknown if the effects we observe in late L3 are a result of only CP-related mechanisms, or are a consequence of the former plus additional homeostatic compensation occurring during larval life. The ability to record from *Drosophila* embryos (Baines and Bate, 1998 [↗](#)), in addition to all larval stages, means that this important question can be addressed experimentally.

Our results show excitatory input to motoneurons is increased to a greater extent than inhibitory input when the developing circuit is exposed to excessive activity during the CP. This could be considered surprising. This is because it may be assumed that the nervous system should compensate for increased excitation by decreasing excitability through either: (1) reducing excitatory (A18a) input strength; (2) increasing inhibitory (A31k) input strength; (3) both 1 and 2. A potential explanation may be that activity during a CP is not compensated for; rather it is used to encode homeostatic set points to which the mature nervous system will continually attempt to return. Set points are defined as ‘ranges of activity’ associated with normal circuit function, so that activity outside of the set point may induce abnormal behaviour (Styr *et al.*, 2019 [↗](#); Truszkowski and Aizenman, 2015 [↗](#); Tien and Kerschensteiner, 2018 [↗](#)). Inappropriate E:I balance during the CP may establish a non-physiologically relevant set point which then serves as the basis for homeostasis once the CP is closed. If CPs capture, rather than compensate for activity to establish set points, it is logical that the strength of premotor inputs increases in accordance with the hyperexcitability induced by exposure to PTX during the CP. It may also explain how the otherwise life-long hyperexcitation caused by the *para*^{bss} mutation can be rescued by inhibiting neural activity during the short 2h CP window (Giachello and Baines, 2015 [↗](#)). Alternatively, whilst the increased inhibition we observe is not statistically significant ($p = 0.15$), it is close and has a medium effect size (Cohen’s $d = 0.78$), and thus may be indicative of an attempt by the locomotor network to rebalance activity back towards a genetically pre-determined level. In this regard, it may just not have sufficient range to be able to counter the increase in excitation due to CP manipulation.

How neurons measure levels of activity during the CP remains under active investigation (Zeng *et al.*, 2021 [↗](#); Carreira-Rosario *et al.*, 2021 [↗](#); Giachello *et al.*, 2021 [↗](#); Coulson *et al.*, 2022 [↗](#); Reh *et al.*, 2020 [↗](#); Peters and Naneix, 2022 [↗](#); Gibel-Russo, Benacom and Di Nardo, 2022 [↗](#)). The second messenger calcium is likely to be involved, as calcium signalling is central to activity-regulated plasticity during nervous system development, including adjustment to changing internal and external environments, as well as learning and memory (Spitzer, 2012 [↗](#)). Similarly, we expect reactive oxygen species to have an important role. Reactive oxygen species have been identified as obligate signals required for activity-regulated synaptic plasticity (Sobrido-Cameán *et al.*, 2023 [↗](#); Oswald *et al.*, 2018 [↗](#); Peng *et al.*, 2019 [↗](#); Milton *et al.*, 2011 [↗](#); Doser and Hoerndli, 2021 [↗](#)). At the same time, dysregulation in reactive oxygen species levels or buffering has been identified as one of the earliest indicators in human CP-associated neurodevelopmental conditions (Steullet *et al.*, 2017 [↗](#)). There are numerous intersections between calcium and reactive oxygen species signalling. For example, mitochondria are well known regulators as well as sensors of intracellular calcium (Rizzuto *et al.*, 2012 [↗](#)) and a recent study has explored how in neurons activity-dependent changes in mitochondrial calcium regulate changes in metabolism, which invariably link to changes in the rate of reactive oxygen species generation (Díaz-García *et al.*, 2021 [↗](#)). We had previously shown that nitric oxide plays an important role during the CP of the

Drosophila embryo, potentially as an activity-regulated signal that mediates adjustment within an interconnected network (Giachello *et al.*, 2021 [DOI](#)). Nitric oxide might also intersect with calcium signalling, as a modifier of CamKII (Araki *et al.*, 2020 [DOI](#)). It will be interesting to explore both calcium and reactive oxygen species signalling further in the context of neuronal plasticity and CPs.

Our observations of lasting change to synaptic physiology provides further evidence to support the hypothesis that physiologically-balanced E:I, during a CP is key to normal network function. Similarly, our results that show that a ch neuron inhibitory “brake” (Carreira-Rosario *et al.*, 2021 [DOI](#)) provides a dose-dependent rescue of excessive excitation caused by PTX or the *para*^{bss} mutation, are consistent with a CP integrating activity to which it is exposed. It is notable that 80db-induced ch activity rescued both the PTX and *para*^{bss} seizure phenotype completely. The simplest explanation for this is that 80dB inhibition matches the strength of both types of increased excitation so they cancel each other out. That is to say, the degree of hyperexcitation caused by PTX and *para*^{bss} are similar. However, this may be an overly simplistic assumption and instead ch neuron activity may achieve rescue via alternate means including, but not limited to, an influence on activity-dependent CPG wave frequency during the CP.

Ch neurons have been implicated in establishing CPG rhythm via connections to premotor and pacemaker neurons A27h and M, respectively (Zeng *et al.*, 2021 [DOI](#)). This rhythm-setting function may be related to relaying early muscle activity and coordinating it with the output of the CNS. Muscles produce myogenic activity and twitch early in development, beginning around 15hrs AEL (Crisp *et al.*, 2008 [DOI](#); Crisp, Evers and Bate, 2011 [DOI](#); Baines and Bate, 1998 [DOI](#)). Given that this is just before the locomotor circuit CP (17-19 h AEL) (Giachello and Baines, 2015 [DOI](#)) it is possible that the development of coordinated activity in the locomotor circuit depends on a succession of set points established in muscles, then peripheral neurons (e.g. ch neurons), central neurons (A18a, A31k, M and A27h neurons) and motor neurons (aCC). Finally, despite the ch brake being inhibitory in the sense that it slows the CPG (Carreira-Rosario *et al.*, 2021 [DOI](#)) and counteracts the excessive excitation provided by PTX or *para*^{bss}, electrophysiological recording from aCC shows that stimulating ch neurons (by optogenetics) is sufficient to depolarise aCC (IH & RAB, unpublished data). The idea that a depolarising input can inhibit activity in the locomotor circuit is likely explained by the contribution ch neurons make to the CPG rhythm. CPG activity causes regular significant depolarisation of aCC in the form of spontaneous rhythmic currents (SRCs). SRCs may be followed by refractory periods that prevent further (i.e. greater) activity occurring in the network. This could be explored in future work.

The rescue of increased seizure in *para*^{bss} reported (Giachello *et al.*, 2015) was achieved using a different modality to inhibit neurons (optogenetics) than we have used here (vibration). The latter is more physiological; it is known that mechanoreceptors provide activity that contributes to normal development of locomotor circuits / CPGs in flies and other animals (Carreira-Rosario *et al.*, 2021 [DOI](#); Zeng *et al.*, 2021 [DOI](#)). Indeed, our validation experiment showed that activating proprioceptors during just the CP, in the absence of PTX, is sufficient to induce a seizure phenotype.

Vibration represents one of a growing list of approaches that show that manipulating neural activity during development can cause permanent change to network function. Other methods for manipulating activity include endogenous mutations (e.g. *para*^{bss}), exposure to drugs (PTX or phenytoin (Giachello and Baines, 2015 [DOI](#))), genetically targeted expression of tetanus toxin (Carreira-Rosario *et al.*, 2021 [DOI](#)), temperature (Kiral *et al.*, 2021 [DOI](#)) and light (via optogenetics (Giachello and Baines, 2015 [DOI](#))). Some (e.g., temperature and vibration) may be encountered by animals developing in the wild, and can be excitatory or inhibitory (for example, altered temperature can increase or decrease neural activity during development). Thus, our work, in combination with that conducted by others, suggests that CPs may ‘measure’ neural activity which is influenced by various environmental and internal factors. This measurement is used to set key

network parameters according to the context in which the circuit functions. In support of this, a recent study reports that synaptic number in the *Drosophila* adult eye scales inversely with developmental temperature (Kiral *et al.*, 2021 [DOI](#)). Behavioural analysis indicates that this phenomenon ensures that adult movement is highly dependent on, and optimized for, developmental temperature.

In conclusion, our work provides novel insight into how activity manipulation, during a defined CP, results in mature network change. Specifically, we show a clear and lasting change to network E:I balance that leaves the locomotor network prone to seizure following a sufficiently large stimulus and relate this to cellular and synaptic-level changes in the circuit. The changes we observed may reflect a network that has encoded a non-physiological set point that, though capable of supporting normal locomotion, is not sufficient to guard against the effects of strong stimulation. Finally, we show that changes induced by excessive excitation can be compensated by inhibition provided by mechanosensory neurons in a dose-dependent manner. This supports the view that a CP acts to tune a developing network based on the influences on activity to which it is exposed.

Materials and Methods

Drosophila Rearing and Stocks

Drosophila stocks were kept on standard corn meal medium, at 25°C. For all experiments where crosses were necessary, stocks were selected and combined as a 1:2 ratio of Gal4-line males to UAS-line virgin females.

Wild-type *Canton-S* or bang-sensitive (a.k.a. *bss*, *para*^{*bss1*}) were from our lab stocks. The following lines were used for GCaMP imaging and/or electrophysiology: *w*; *GMR94G06-Gal4* (aCC, gift from Matthias Landgraf); *w*; *15B07-Gal4* (A18a, a.k.a. A18a-Gal4, gift from Aref Zarin); *R20A03-p65ADzp in attP40*; *R93B07-ZpGDBD in attP2* (A31k, a.k.a. A31k-Gal4, gift from Akinao Nose); *P{iav-GAL4.K}3* (chordotonal neurons, BDSC #52273); *w[*]; P{y[+t7.7] w[+mC]=20XUAS-Chronos::mVenus}attP40* (BDSC #77115); *w[1118]; P{y[+t7.7] w[+mC]=20XUAS-IVS-jGCaMP7f}su(Hw)attP5* (BDSC #80906); *w*; *13xLexAop2-Flp2@attP40, Drep2^{FonYPet}, 13xLexAop2-IVS-myr::tdTom @ su(HW)attP5/CyO, Dfd-GMR::YFP; RN2-FLP-hopA, tub-FRT-stop-FRT-LexAVP16 (No.5)/TM6b, Sb, Dfd-GMR::YFP* (a.k.a. RN2-FLP LexA-tdTom Drep2::YPet) for microscopy.

Manipulation of Activity during Development

PTX feeding protocol for increasing neural activity during development: PTX and vehicle food were made by pipetting 20µl 25mg/ml PTX, or an equivalent volume of vehicle (ethanol, EtOH) to 1ml of ddH₂O, for a final concentration of 415µM PTX. 4g yeast extract powder (Melford, UK) was stirred into the 1ml ddH₂O to make 5g yeast paste food. ~1ml of either PTX or control food was then placed on the surface of a 6ml grape agar plate. Flies were added to a 50ml beaker using a funnel, and the beaker was connected to a grape agar plate (with food) using a pressure-fit plastic collar, to form a small laying pot. Laying pots were left at 18°C for 48h (over weekend) to allow flies to acclimate and eat the food. After 48h, pots were moved to 25°C and changed (beaker swapped and food replenished) at 10:00 and 15:00 every day, for five days (Monday-Friday). Embryos were collected from the surface of grape agar plates, using a wet paintbrush, for the final three days (Wednesday-Friday) of this process. Following collection, embryos (of either sex) were gently placed on the surface of vials containing either: (1) corn meal medium enriched with ~70µM all-trans retinal (Sigma-Aldrich, UK, 50µl of 100mM stock pipetted onto ~7ml food and left to dry before embryos were added), a cofactor chromophore necessary for activation of channelrhodopsin, for animals to be used in optogenetic experiments; (2) normal corn meal medium, in the case of embryos developing into larvae used in GCaMP imaging or behavioural experiments.

Vibration protocol for activating chordotonal neurons during embryonic development

Gravid females were fed either: (1) normal corn meal medium where effect of chordotonal neurons was not used to rescue effect of PTX (e.g. *Canton-S* and *para^{bss}* experiments); (2) 830µm PTX food (as above), when chordotonal neuron activity was used to rescue effect of PTX. Male and female embryos were collected Monday-Thursday and timed precisely (from animals laying in a four-hour window between 10:30-14:30), to ensure that the developmental stage at which embryos were exposed to vibration included the 17-19h after egg laying (AEL) CP. Embryos were collected onto grape agar plates and arranged dorsal side up. Plates were then placed on a custom-made perspex shelf glued to a speaker (Adafruit STEMMMA Speaker - Plug and Play Audio Amplifier), in a 25°C incubator. The speaker was wired to an Arduino Uno Rev 3 printed circuit board with an ATmega328P microcontroller, that was programmed using C++, via the Arduino IDE (Arduino 1.8.19), using the following code: <https://github.com/virhunter/vibrationduringdevelopment/blob/main/timedbuzzer> .

The code was written and refined by several users of the Arduino forum. The original forum thread, which includes the usernames of those who contributed to it, is visible at: <https://forum.arduino.cc/t/help-with-basic-buzzer-tone-program-will-acknowledge-in-scientific-journal-article-in-return/1027170/25> . The code instructed the Arduino to power the speakers to provide a 1kHz tone (vibration) to the perspex shelf (so the embryos), with an on-off frequency of 0.5Hz (ON 1s, OFF 1s) for 6 hours, starting 12.75h after a button was pressed. Thus, once embryos were collected and placed on the shelf at 14:30, the button was pressed at 14:45, and stimulation was delivered to embryos between 03:30 and 09:30. Tone volume was 80dB for all experiments besides those where others are stated explicitly (e.g., 40 and 60dB).

Crawling velocity

Male and female wall-climbing third instar larvae (L3) were selected from vials and washed to remove food residue. They were dried with a tissue and placed in the centre of a 1% agarose gel arena surrounded by a 5M NaCl (Sigma-Aldrich) moat. Animals were then left to acclimatise to the arena and recover from handling, for 30 seconds. Crawling was then recorded for 120s, using a DanioVision observation chamber (Noldus, Netherlands). Only larvae that spent 100% of recording time within the gel arena were included in analysis (the few that entered the moat, were excluded). Velocity was calculated from movement of a centroid automatically identified by the tracking system, in mm/s, by the Ethovision software (Noldus, v. XT 11).

Recovery from electroshock

Assays were conducted as described previously (Marley and Baines, 2011a ). In brief, male and female wall-climbing L3 larvae were selected from vials and washed to remove food residue. L3 were dried with a tissue and placed on an plastic plate lid to recover from handling. Once animals had recovered and normal crawling had resumed, a handheld stimulator composed of two tungsten wires fixed to a nonconducting rod (tips of the two wires (0.1 mm diameter) were set ~1–2 mm apart) was positioned and very gently pressed into the larval cuticle, perpendicular to and over the approximate position of the CNS. A direct current pulse (10V, 1s) generated by a DS2A Mk.II stimulator (Digitimer Ltd., UK) was delivered via the probe to the animals, which usually responded with a sustained contraction and subsequent immobility representing seizure-like behaviour. The time to resumption of normal behaviour (regular, whole body peristaltic waves that produce forward locomotion) was recorded and defined as duration of recovery from seizure. Strong peripheral stimulation likely causes excessive and synchronous synaptic excitation within the CNS resulting in seizure. However, the precise mechanism of this effect remains to be determined.

Electrophysiology

Recordings were performed as described previously (Baines et al., 1990; Marley and Baines, 2011), in male and female L3 larvae. In brief, the protocol was as follows: larvae were dissected in a dish in saline (135mM NaCl (Fisher Scientific, UK), 5mM KCl (Fisher Scientific), 4mM MgCl₂·6H₂O (Sigma-Aldrich), 2mM CaCl₂·2H₂O (Fisher Scientific), 5mM TES (Sigma-Aldrich), 36mM sucrose (Fisher Scientific), at pH 7.15) to remove the CNS. The isolated CNS preparation was then transferred to a 200µl droplet of external saline, within which it was laid flat (dorsal side up) and glued to a Sylgard-coated cover slip on a slide. 200µM mecamylamine (Sigma-Aldrich) was added to the droplet of saline for recordings made following optogenetic activation of A31k, but was not added for those made following activation of cholinergic A18a.

Before recording, interneuron Gal4 and mVenus-tagged reporter line expression (e.g. 15B07-Gal4 x UAS-Chronos::mVenus) was checked by momentarily exposing preparations to blue light (470nm OptoLED, Cairn Instruments, UK), while viewing them under a microscope (model BX51WI, Olympus, UK). Once expression was confirmed, aCC motoneurons present in the VNC segments that expressed the interneuron Gal4 and reporter, were identified using bright-field microscopy. 1% protease (Sigma-Aldrich) was applied to those segments to remove overlying glia, to facilitate access to aCC. Recordings were made using borosilicate glass pipettes (GC100F-10, Harvard Apparatus, USA) that were fire polished to resistances of 10–15MΩ and filled with intracellular saline (140 mM K1-D-gluconate (Sigma-Aldrich), 2 mM MgCl₂ · 6H₂O (Sigma-Aldrich), 2 mM EGTA (Sigma-Aldrich), 5 mM KCl (Fisher Scientific), and 20 mM HEPES (Sigma-Aldrich), at pH 7.4). Input resistance was measured in the ‘whole cell’ conformation, and only cells that: (1) had an input resistance ≥0.5GΩ; (2) had a membrane potential ≤−40 mV; (3) were not firing action potentials at rest, were used for experiments. Data was captured using a Multiclamp 700B amplifier controlled by pCLAMP (v. 11.2.1), via a analog-to-digital converter (model Digidata 1440A, Molecular Devices, CA, USA). Traces were sampled at 20 kHz and filtered online at 10 kHz.

Current-clamp recordings from aCC were made by injecting current (−5–10 pA) sufficient to evoke action potentials (APs) in aCC, at a frequency of ~5–10 Hz. Premotor synaptic drive was elicited by optogenetic stimulation of interneurons using green light (565nm OptoLED, Cairn Instruments) controlled by a Clampex (version 10.7) protocol comprised of three sweeps (repetitions) of 1s LED off, 1s on, 1s off, for each preparation. Change in frequency (ΔAP Freq. (Hz)) was calculated as the difference in number of action potentials firing before and during optogenetic stimulation.

GCaMP Imaging

Imaging was performed on aCC motor neurons in w; UAS-GCaMP7f x w;;GMR94GO-Gal4 male and female L3 derived from gravid females fed PTX, or vehicle control (see *Manipulation of Activity during Development*). larvae were dissected in a dish in external saline (see *Electrophysiology*) to remove the CNS. The isolated CNS preparation was then transferred to a droplet of saline, within which it was laid flat (dorsal side up) on a Sylgard-coated cover slip on a slide. The slide was positioned under a microscope (model BX51WI, Olympus) with a Hamamatsu ORCA-flash 4.0 C114440 digital camera, and the CNS viewed using a x20 water immersion lens. Blue light (a 470nm OptoLED, Cairn Instruments) was used to excite GCaMP, and fluorescence was recorded for 180s at a 10Hz imaging rate, using Winflow v4.1.3 (University of Strathclyde). During analysis, ROIs were placed on aCC axons, on the same side of the VNC, from A6–A4, and exported into Clampex (10.7). Traces were smoothed (add type and amount), with forward wave propagation being measured between adjacent ipsilateral ROIs. Time lag was measured as the mean duration between events (peak amplitudes), between segments A6–A5 and A5–A4. Means were calculated from up to 5 individual measures of time lag.

Confocal Microscopy

PTX feeding protocol

Picrotoxin (Merck, NJ, USA) was prepared to 25mg/ ml in 100% ethanol, then 40ul of this stock diluted 50-fold by adding to 1960ul ddH₂O then mixed with 8g of yeast extract powder (Melford), while for controls the equivalent amount of EtOH only was used. Canton-S virgin females were crossed to RN2-FLP LexA-tdTom Drep2::YPet males (see above for complete genotype) and left to mate in vials at 25°C for 24 hours. Mated flies were distributed to two laying pots with apple juice agar plates, with an excess (~0.5g) of either picrotoxin or control yeast extract paste. Laying pots were kept at 25°C, yeast plates changed and discarded daily for three days to ensure sufficient time for drug exposure. Thereafter, newly hatched larvae aged to within 1 hour of one another were picked from laying plates and transferred to new apple juice agar plates with an excess of normal yeast paste (dried baker's yeast and ddH₂O), and allowed to develop at 25°C for 72 hours.

Imaging

At 72 hours after larval hatching (ALH), larvae of the correct genotype were chosen by selection against the Dfd-GMR-YFP balancer. Larval nerve cords were dissected at room temperature in Sorensen's saline, adhered to a poly-L-lysine (Merck) coated coverglass, and imaged 'live' within a drop of Sorensen's saline on a Leica Stellaris 5 upright confocal microscope with a Leica 506148 HC APO L 63x/0.9 W U-V-I Water Immersion objective. The following imaging settings were used: format 800×800, 16bit, scan speed 400Hz, bidirectional scanning, pinhole 1 Airy Unit, z step size 0.3um, sequential frame scanning. For Drep2::YFP detection: excitation 514nm at 1.5% power, gain 20, emission collection window 520-540nm, frame accumulation 3. For myr::tdTom detection: excitation 561nm at 2% power, gain 15, emission collection window 570-700nm, line accumulation 4. Inbuilt Leica LIGHTNING deconvolution software was used, with default settings optimised for water immersion. Following acquisition of the 'live' image, nerve cords were fixed for 15 minutes at room temperature in 2% PFA in Sorensen's saline, washed 3x in Sorensen's saline, blocked for 1 hour at 4°C in blocking solution (1x Sorensen's saline, 0.3% Triton-X-100, 0.25% BSA) and then incubated for 15 hours at 4°C with FluoTag-X4 anti-RFP STAR635P and FluoTag-X4 anti-GFP ATTO488 (NanoTag Biotechnologies), both diluted 1:500 in blocking solution. Stained samples were then washed 3x in Sorensen's saline at 4°C for >1 hour, before imaging again. Imaging settings for nanobody-stained samples were the same as for live samples, with the following specific differences: format 1024×1024, z step size 0.25um. For Drep2::YFP > anti-GFP ATTO488 detection: excitation 488nm at 1.5% power, gain 2.5, emission collection window 498-580nm, line averaging 3, line accumulation 3. For myr::tdTom > anti-RFP STAR635P detection: excitation 638nm at 2% power, gain 15, emission collection window 648-830nm, line averaging 3, line accumulation 3.

Image analysis

Imaris v10.0 software (Bitplane) was used for image analysis. Dendrites were reconstructed from the myr::tdTom channel of 'live images' as dendrite morphology is clearer pre-fixation. The Imaris filaments module was used for dendrite reconstruction, using the tree autopath algorithm, with the starting points threshold set to include just one starting point on the cell body. Seed points for segments multiscale points were set to a thinnest diameter of 0.01um, and largest diameter of 3um. Iterative rounds of user-informed AI training were used for seed point classification until all seed points correctly coincided with dendrites, followed by iterative rounds of AI training for segment classification. A tree-build maximum gap length of 5um was used. Drep2::YPet marking post-synaptic sites was detected from the fixed and nanobody stained images, since Drep2::YPet>anti-GFP ATTO488 showed good photostability through the z-stack compared with 'live' YPet protein, which suffered substantial photobleaching through the course of acquiring a 'live' image stack. The Imaris spots module was used for Drep2::YPet spot detection, using an estimated xy diameter of 0.1um, and z diameter of 0.5um. The spot quality threshold was

informed by the frequency histogram and set to the trough point, at which any further decrease in threshold results in a sharp increase in spot number, corresponding to false positive spots detected outside the cell from low-level background. In this way, all Drep2::YPet puncta of different sizes were detected, while avoiding false positives from low level background.

Statistical Analysis

Data were imported into and organised in Microsoft Excel (Microsoft Corporation, WA, USA), and GCaMP imaging activity peaks were smoothed and quantified in ClampFit, before being copied into GraphPad Prism 9 (GraphPad Software) for statistical analysis. Normality was tested using the Shapiro-Wilk test, with further analysis conducted according to the nature of the distribution. Specifically, differences between two groups of normally distributed data for crawling velocity, recovery time from electroshock, change in aCC evoked action potential firing frequency in response to premotor interneuron stimulation and mini amplitude or frequency, were analysed by student's t-tests and are shown as mean $\pm$ SEM. Normally distributed data compared across three groups (e.g., all experiments that showed rescue of activity imbalance by ch stimulation, besides synchronicity (see below) was analysed using one-way ANOVA with multiple comparisons. Data for rescue of synchronicity following exposure to PTX by ch stimulation were not normally distributed, so were analysed using one-way ANOVA with Kruskal-Wallis multiple comparisons test. p values < 0.05 were considered significant, and levels of significance were represented by: * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$.

Effect size (Cohen's d) for change in aCC evoked action potential firing frequency in response to premotor interneuron stimulation was calculated manually, using the formula: $Cohen's d = (M_1 - M_2)/SD_{pooled}$, where: M_1 = mean aCC action potential firing frequency before activation of premotor neuron; M_2 = mean aCC action potential firing frequency during activation of premotor neuron; $SD_{pooled} = \sqrt{((SD_1^2 + SD_2^2)/2)}$.

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Competing Interests

The authors declare no competing interests.

Author Contributions Statement

I.H., B.C., T.P., M.L. and R.A.B. designed research; I.H., B.C., T.P., J.J.D. and J.P. performed research; I.H., B.C., T.P., J.J.D. and J.P. analysed data; I.H. wrote the manuscript; M.L. and R.A.B. provided feedback on and edited drafts.

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Editors

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Reviewer #1 (Public Review):

Activity has effects on the development of neural circuitry during almost any step of neuronal differentiation. In particular during specific time periods of circuit development, so called critical periods (CP), altered neural activity can induce permanent changes of neuronal and network excitability. In complex neural networks it is often difficult to pinpoint the

specific network components that are permanently altered by activity, and it often remains unclear how activity is integrated during the CP to set mature network excitability. This study combines electrophysiology with pharmacological and optogenetic manipulation in the *Drosophila* genetic model system to pinpoint the neural substrate that is influenced by altered activity during a critical period (CP) of larval locomotor circuit development. Moreover, it is then tested whether and how different manipulations of synaptic input are integrated during the CP to tune network excitability.

Strengths: Based on previous work, during the CP network activity is increased by feeding the GABA-AR antagonist PTX. This results in permanent network activity changes as highly convincingly assayed by a prolonged recovery period following induced seizure and by altered intersegmental locomotor network coordination. This is then used to provide two important findings: First, compelling electro- and optophysiological as well as anatomical experiments track the site of network change down to the level of single neurons and pre-versus postsynaptic specializations. In short, increased activity during the CP increases both, the magnitude of excitatory and inhibitory synaptic transmission to the aCC motoneuron, but excitation is affected more strongly. This results in altered excitation inhibition ratios. Fine electrophysiology shows that excitatory synapse strengthening occurs postsynaptically. High quality anatomy shows that dendrite size and numbers of synaptic contacts remain unaltered. It is a major accomplishment to track the tuning of network excitability during the CP down to the physiology of specific synapses at identified neurons.

Second, additional experiments with single neuron resolution demonstrate that during the CP different forms of activity manipulation are integrated so that opposing manipulations can rescue altered setpoints. This provides novel insight into how developing neural network excitability is tuned, and it indicates that during the CP training can rescue the effects of hyperactivity.

Weaknesses: There are no major weaknesses to the findings presented, but the molecular cause that underlies increased motoneuron postsynaptic responsiveness as well as the mechanism that integrates different forms of activity during the CP remain unknown. However, the discussion addresses this point adequately.

Reviewer #2 (Public Review):

SUMMARY: In this study, the authors use the tractable *Drosophila* embryonic/larval motor circuit to determine how manipulations to activity during a critical period (CP) modify the circuit in ways that persist into later developmental stages. Previously, this group demonstrated that manipulations to the aCC/MN-Ib neuron in embryonic stages enhance (or can rescue) susceptibility to seizures at later larval stages. Here, the authors demonstrate that following enhanced excitatory drive (by PTX feeding), the aCC neuron acquires increased sensitivity to cholinergic excitatory transmission, presumably due to increased postsynaptic receptor abundance and/or sensitivity, although this is not clarified. Although locomotion is not altered at later developmental larval stages, the authors suggest there is reduced "robustness" to induced seizures. The second part of the study then goes on to enhance inhibition during the CP in an attempt to counteract the enhanced excitation, and show that many aspects of the CP plasticity are rescued. The authors conclude that "average" E/I activity is integrated during the CP to determine excitability of the mature locomotor network.

Overall, this study provides compelling mechanistic insight into how a final motor output neuron changes in response to enhanced excitatory drive during a CP to change functionality of the circuit at later mature developmental stages. The first part of this study is strong, clearly showing the changes in the aCC neuron that result from enhanced excitatory input. This includes very nice electrophysiology and imaging data that assess synaptic function and structure onto aCC neurons from pre-motor inputs resulting from PTX exposure during development. However, the later experiments in Figures 6 and 7 designed to counteract the CP plasticity are somewhat difficult to interpret. In particular, the specificity of the

manipulations of the ch neuron intended to counteract the CP plasticity is unclear, given the complexities of how these changes impact excitability all neurons during development. It is clear that CP plasticity is largely rescued in later stages, but it is hard to know if downstream or secondary adaptations may be masking the PTX-induced plasticity normally observed. Nonetheless, this study provides an important advance in our understanding of what parameters change during CPs to calibrate network dynamics at later developmental stages.

Reviewer #3 (Public Review):

Summary:

In Hunter, Coulson et al, the authors seek to expand our understanding of how neural activity during developmental critical periods might control the function of the nervous system later in life. To achieve increased excitation, the authors build on their previous results and apply picrotoxin 17-19 hours after egg-laying, which is a critical period of nervous system development. This early enhancement of excitation leads to multiple effects in third-instar larvae, including prolonged recovery from electroshock, increased synchronization of motor neuron networks, and increased AP firing frequency. Using optogenetics and whole-cell patch clamp electrophysiology, the authors elegantly show that picrotoxin-induced over-excitation leads to increased strength of excitatory inputs, and not loss of inhibitory inputs. To enhance inhibition, the authors chose an approach that involved stimulation of mechanosensory neurons; this counteracts picrotoxin-induced signs of increased excitation. This approach to enhancing inhibition requires further validation.

Strengths:

- The authors confirm their previous results and show that 17-19 hours after egg laying is a critical period of nervous system development.
- Using $\text{Ca}^{2+}/\text{Sr}^{2+}$ substitutions, the authors demonstrate that synaptic connections between A18a & aCC show increased mEPSP amplitudes. The authors show that this aCC input is what is driving enhanced excitation.
- The authors demonstrate that the effects of over-excitation attributed to picrotoxin exposure are generalizable and also occur in bss mutant flies.

Weaknesses:

- The authors build on their previous work and argue that the critical period (17-19h after egg-laying) is a uniquely sensitive period of development. Establishing the developmental window of the critical period is important for the present study. The present study would benefit from demonstrating that exposure to picrotoxin at L1 or L2 do not lead to changes in induced seizure at L3. This would further the authors hypothesis of the criticality of the 17-19h AEL period.
- The ch-related experiments require further controls and explanation. Regarding experiments in Fig 6, what is the effect of ch neuron stimulation alone on time lag and AP frequency? The authors report related pilot experiments have been performed; the present study would be strengthened with inclusion of these data.

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Activity has effects on the development of neural circuitry during almost any step of differentiation. In particular during specific time periods of circuit development, so-called critical periods (CP), altered neural activity can induce permanent changes in network excitability. In complex neural networks, it is often difficult to pinpoint the specific network components that are permanently altered by activity, and it often remains

unclear how activity is integrated during the CP to set mature network excitability. This study combines electrophysiology with pharmacological and optogenetic manipulation in the *Drosophila* genetic model system to pinpoint the neural substrate that is influenced by altered activity during a critical period (CP) of larval locomotor circuit development. Moreover, it is then tested whether and how different manipulations of synaptic input are integrated during the CP to tune network excitability.

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Second, additional experiments with single neuron resolution demonstrate that during the CP different forms of activity manipulation are integrated so that opposing manipulations can rescue altered setpoints. This provides novel insight into how developing neural network excitability is tuned, and it indicates that during the CP, training can rescue the effects of hyperactivity.

Weaknesses:

There are no major weaknesses to the findings presented, but the molecular cause that underlies increased motoneuron postsynaptic responsiveness as well as the mechanism that integrates different forms of activity during the CP remain unknown. It is clear that addressing these experimentally is beyond the scope of this study, but some discussion about different candidates would be helpful.

We discuss likely mechanisms that underpin the increase in postsynaptic responsiveness below (Reviewer #1 (Recommendations For The Authors);, point 2). To address possible mechanisms that integrate different forms of activity we now include a new paragraph in the discussion.

Reviewer #2 (Public Review):

Summary:

In this study, the authors use the tractable *Drosophila* embryonic/larval motor circuit to determine how manipulations of activity during a critical period (CP) modify the circuit in ways that persist into later developmental stages. Previously, this group demonstrated that manipulations to the aCC/MN-Ib neuron in embryonic stages enhance (or can rescue) susceptibility to seizures at later larval stages. Here, the authors demonstrate that following enhanced excitatory drive (by PTX feeding), the aCC neuron acquires increased sensitivity to cholinergic excitatory transmission, presumably due to increased postsynaptic receptor abundance and/or sensitivity, although this is not clarified. Although locomotion is not altered at later developmental larval stages, the authors

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In Hunter, Coulson et al, the authors seek to expand our understanding of how neural activity during developmental critical periods might control the function of the nervous system later in life. To achieve increased excitation, the authors build on their previous results and apply picrotoxin 17-19 hours after egg-laying, which is a critical period of nervous system development. This early enhancement of excitation leads to multiple effects in third-instar larvae, including prolonged recovery from electroshock, increased synchronization of motor neuron networks, and increased AP firing frequency. Using optogenetics and whole-cell patch clamp electrophysiology, the authors elegantly show that picrotoxin-induced over-excitation leads to increased strength of excitatory inputs and not loss of inhibitory inputs. To enhance inhibition, the authors chose an approach that involved the stimulation of mechanosensory neurons; this counteracts picrotoxin-induced signs of increased excitation. This approach to enhancing inhibition requires further control experiments and validation.

Strengths:

- The authors confirm their previous results and show that 17-19 hours after egg laying is a critical period of nervous system development.
- Using $\text{Ca}^{2+}/\text{Sr}^{2+}$ substitutions, the authors demonstrate that synaptic connections between A18a $\rightarrow$ aCC show increased mEPSP amplitudes. The authors show that this aCC input is what is driving enhanced excitation.
- The authors demonstrate that the effects of over-excitation attributed to picrotoxin exposure are generalizable and also occur in bss mutant flies.

Weaknesses:

- The authors build on their previous work and argue that the critical period (17-19h after egg-laying) is a uniquely sensitive period of development. Have the authors already demonstrated that exposure to picrotoxin at L1 or L2 (and even early L3 if experimentally

possible) does not lead to changes in induced seizure at L3? This would further the authors' hypothesis of the uniqueness of the 17-19h AEL period. If this has already been established in prior publications, then this needs to be further explained. I do note in Gaicelllo and Baines (2015) that Fig 2E shows the identification of the 17-19h window.

This is a pertinent comment. We now have evidence that activity manipulation (in this instance by increasing temperature, which recapitulates the effect of PTX) is not effective at larval stages (L1 to L3) but remains effective between 17-19hrs AEL. This observation forms part of a separate study where we explore the role of circadian activity on embryonic and larval neuronal development. We include a brief statement to address this comment in the revision (first paragraph of Results).

• Regarding experiments in Fig 2, authors only report changes in AP firing frequency. Can the authors also report other metrics of excitability, including measures of intrinsic excitability with and without picrotoxin exposure (including RMP, Rm)? Was a different amount of current injection needed to evoke stable 5-10 Hz firing with and without picrotoxin? In the representative figure (Fig. 2A), it appears that the baseline firing frequencies are different prior to optogenetic stimulation.

No differences in RM, Rin or capacitance were observed due to PTX. This is now included in the revision along with an explanation that different levels of current injection were used to measure effects to excitatory vs inhibitory synaptic drive. We did not specifically monitor the amount of current required to maintain stable firing.

• The ch-related experiments require further controls and explanation. Regarding experiments in Fig 6, what is the effect of ch neuron stimulation alone on time lag and AP frequency? Can the authors further clarify what is known about connections between aCC and ch neurons? It is difficult for this reviewer to conceptualize how enhancing ch-mediated inhibition would worsen seizures. While the cited study (Carreira-Rosario et al 2021) convincingly shows that inhibition of mechanosensory input leads to excessive spontaneous network activity, has it been shown that the converse - stimulation of ch neurons - indeed enhances network inhibition?

• The interpretation of ch-related experiments is further complicated by the explanation in the Discussion that ch neuron stimulation depolarizes aCC neurons; this seems to undercut the authors' previous explanation that the increased E:I ratio is corrected by enhanced inhibition from ch neurons. The idea that ch neurons are placing neurons in a depolarized refractory state is not substantiated by data in the paper or citations.

To respond to these two points combined: The reviewer is correct in stating that additional experiments will be required to fully understand mechanism. We believe that cholinergic (excitatory) chordotonal input to aCC may be an important component for setting the rhythm of the locomotor CPG. Indeed, it may be that CPG rhythm is a key factor during the CP. Our observations suggest optogenetic stimulation of Ch neurons alone is sufficient to induce large, ~400-, currents that resemble endogenous spontaneous rhythmic currents (SRCs) associated with CPG activity. SRCs occur with a characteristic frequency of ~1Hz, and we have some unpublished data that suggests it is possible to change this frequency using ch stimulation. This data therefore unifies prior work (Carreira-Rosario et al., 2021 description of a brake) with our own (observation that ch depolarize aCC). However, we do not include this speculation in the Discussion because the experiments we have conducted were pilots. They may be expanded upon and included in future work.

• In the Discussion, the authors suggest that enhanced proprioception leading to seizures is reminiscent of neurological conditions. This seems to be an oversimplification.

Connecting abnormal proprioception to seizures is quite different from connecting abnormal proprioception to disorders of coordination. This should be revised.

Because this is peripheral to our main study, we have deleted this from the revision.

Reviewer #1 (Recommendations For The Authors):

1. Although the authors have to be commended for the scrutiny with which they pinpoint a site of circuit change, it cannot be excluded that other parts of the circuit also undergo adjustments in response to activity manipulation during the CP, e.g. the membrane properties of the interneurons. This is not a problem but should be discussed.

We agree with this comment and have added the following text to the discussion.....'However, we recognise that other parts of the locomotor network may also undergo change due to CP manipulation. The advantage of this system is that most of these elements are now open to specific manipulation through cell-specific genetic drivers'. (Discussion paragraph 3)

1. It is surprising that there is no discussion of the potential molecular cause for the observed increases in postsynaptic responses to SV release from cholinergic neurons. Given that there are no differences in postsynaptic structure, puncta number etc., the subunit composition of the nAChR seems an obvious guess. What is known about the nAChRs subunit composition on aCC, and when during development do the receptors/different subunits become expressed? A paragraph in the discussion on this issue would be highly relevant to the manuscript.

Our own work (unpublished) together with a recent paper from the Littleton lab (<https://www.sciencedirect.com/science/article/pii/S0896627323005810?via%3Dihub#mmc2>) suggests that aCC expresses the majority, if not all, of the 7 alpha and 3 beta subunits that comprise nAChRs. The situation is further complicated by the fact that these receptors are pentameric and are composed of various subunits – the composition significantly altering channel kinetics. Less is known about expression timelines for each receptor subunit, and certainly not in aCC. We already include the following sentence in the results text.....'A change in the frequency of mini excitatory postsynaptic potentials (mEPSPs, a.k.a. minis) would suggest the adaptation is primarily presynaptic (e.g. increased probability of release), whilst a change in distribution and/or amplitude of minis is more consistent with a mechanism acting postsynaptically (e.g. increased or altered receptor subunits).' Given that we know next to nothing about the nAChR subunit composition in aCC and how this might change due to CP manipulation, we feel it better not to speculate further. To help the reader, we include the following sentence in the discussion.....'The precise mechanism contributing to increased mini amplitude remains to be determined, but a plausible scenario may involve change in cholinergic subunit composition.' (Discussion paragraph 3)

1. It would be important to provide the p-values for Figures 1B and C, especially because it seems that the inhibition also becomes stronger upon PTX treatment during the CP. There is no statistical testing mentioned, was no test done or was it not significant? It is agreed that the effect size is clearly stronger for the increased excitation than for the increased inhibition, but looking at the data suggests that the effect on excitation is not much more significant than the effect on inhibition.

The reviewer is referring to Fig 2B&C. P values have been added to both main text and to the figure legend.

1. Associated with the point above, in the discussion line 407 and below the authors come back to this point and reason that it is surprising that increased excitation is not compensated for by homeostatic mechanisms. It is concluded that homeostatic compensation brings the system back to a setpoint that is defined during the critical period, but the setpoint is set higher in this case. However, an alternative explanation is that GABA administration during the critical period causes the excitation set point to be too high, but this is then partially counteracted in a homeostatic manner by increasing inhibition. If the *p*-values in Figures 2B and C are rather similar, this might even be the favorable interpretation.

We believe the reviewer means ‘PTX administration’ and not GABA. This is an interesting idea and one we had not really considered. We address this comment by adding the following text..... ‘Alternatively, whilst the increased inhibition we observe is not statistically significant ($p = 0.15$), it is close and has a medium effect size (Cohen’s $d = 0.78$), and thus may be indicative of an attempt by the locomotor network to rebalance activity back towards a genetically pre-determined level. In this regard, it may just not have sufficient range to be able to counter the increase in excitation due to CP manipulation.’ (Discussion paragraph 5)

1. To assess the magnitudes of A18a-mediated excitation and A31k-mediated inhibition to aCC, changes in aCC firing frequency were measured. For this aCC was injected with current to fire at all. However, the current injections were chosen to cause firing at 5-10 Hz. During a crawling burst, aCC fires well above 100Hz (Kadas et al., 2017). Are the effects also visible at such firing frequencies, or at least across different firing frequencies? I am not asking for additional experiments, but maybe the data are there and can be referred to?

Spiking in aCC occurs as burst firing, evoked by cholinergic synaptic drive, that lasts for ~300ms and achieving firing frequencies of between 50-100Hz (Kadas et al., 2017 and our own unpublished data). We did not test for effects to excitation or inhibition at these higher frequencies. We now make this explicit in the discussion by adding the following sentence..... ‘The firing frequencies that we imposed (1-10Hz) are also lower than seen during fictive locomotion (Kadas et al., 2017), which shows burst firing lasting for ~300 ms and achieving spike frequencies of up to 100Hz.’ (Discussion paragraph 3)

1. In Figure 3B some minis are demarked by green arrows and others are not. Were the non-marked ones not included in the analysis, and what were the criteria to mark some and others not? This is particularly important because the cumulative distribution of minis is analyzed in Figure 3D, and this depends crucially on what qualifies as mini and what does not.

All mini’s are marked by green arrows. The events not marked are not mini’s. *Drosophila* neurons are small and have an unfavourable dendritic structure for recording minis. Thus, we carefully analyse traces by eye taking only events that show very rapid rise times and slower, exponential decay (the typical mini shape). There are, however, other events which are most likely single/multiple channel openings, which due to filtering are rounded. We now include this same trace, greatly expanded, as Fig S1D to show how we identified minis from non-minis.

1. *The asynchronous release experiment under Sr²⁺ seems an elegant way to analyze minis upon optogenetic stimulation of an identified presynaptic cholinergic neuron. I suggest being a little more conservative with the term asynchronous release (or replacing it), which is usually the release of many single vesicles that follow AP-mediated synaptic transmission and has nicely been demonstrated at the Drosophila NMJ (Besse et al., 2007). Also, please show the trace in Figure S2A under Sr²⁺ at a higher pA magnification, it is really hard to see the minis there.*

We have adopted a previously published technique that, in our view, correctly uses the term ‘asynchronous release’. This is not to say that all asynchronous release occurs via the same mechanism. Indeed, the papers that report the technique we use predate Besse 2007. We also expand the trace in Fig S1A (not S2A as wrongly indicated).

Reviewer #2 (Recommendations For The Authors):

1. *Can the authors explain what they think is the parameter of "activity" being measured in the locomotor circuit (mainly aCC) during the CP? Is the aCC neuron simply summing (perhaps through a proxy like Ca²⁺) total excitation/inhibition over time during the CP?*

Reviewer #1 also requests that we discuss how activity is ‘measured’ and thus we now include a dedicated paragraph in the discussion to address this concern. Whether aCC sums ‘average’ activity or perhaps is influenced by activity extremes remains uncertain. Our data is consistent with the former but further work is required to validate our conclusion. This work will be published in due course.

Related to understanding this concept, could the authors' silence activity (using Kir2.1, TNT, or BoNT) from each of the monosynaptic premotor inputs in otherwise wildtype and following PTX exposure to determine how the circuit responds when each of the monosynaptic inputs are silenced? This might inform the role they play in instructing how activity is measured over time during the CP.

This is an excellent suggestion and, indeed, we have planned such experiments. Silencing specific neurons, whilst manipulating the CP, may well result in more significant network instability due to the setting of multiple (and physiologically inappropriate) homeostatic set points. Such studies go beyond the scope of the present study and thus we prefer not to speculate at this early stage, but to wait for experimental data.

On a related note, the authors focus on just 2 premotor inputs, presumably due to the availability of specific drivers. But do the authors know how many other inputs (other ACh, Gaba, and glutamate) onto aCC there are, and to what extent do the authors think these are changed in similar or distinct ways? Is it implied that all neurons are similarly altered by the manipulations?

The connectome details the number and types of neurons that directly contact the aCC motoneuron (Zarin et al., 2019). In terms of cholinergic excitors, the results present in Figure 3 suggest that most (all?) inputs are strengthened following embryonic PTX exposure. However, to conclude this would be highly speculative and thus we refrain from doing so in the manuscript. As other single-neuron driver lines become available, such expts will hopefully be possible.

1. *If PTX treatment does indeed increase CPG synchronicity, shouldn't there be a readout of this effect on larval locomotion? While the speed of locomotion wasn't significantly impacted, perhaps another parameter was altered.*

It is quite possible that other aspects of locomotion are being altered (turning, rearing, etc), but we have not analysed for these more subtle behaviours. Indeed, although not statistically significant, there is a modest reduction in average velocity in larvae derived from PTX-exposed embryos. We see similar reductions in characterised seizure mutants which also show increased synchronicity (Streit et al., 2016).

1. *In Figure 2 and elsewhere, what is the baseline level of AP firing rate in each aCC neuron, before optogenetic stimulation? Is this informative about how PTX exposure alters excitability to begin with, perhaps by changing intrinsic excitability.*

We now include this data in the relevant results section. Interestingly, following exposure to PTX, basal firing was significantly increased in A18a (excitatory premotor) but not in A31k (inhibitory premotor). This reflects our experiment in which we conclude that excitatory drive to aCC is increased relative to inhibitory synaptic drive. Thus, this measure seemingly validates our conclusion that E:I balance has been altered following activity-manipulation during the CP.

1. *Figure 3: The apparent increase in mini amplitude is very small (4.1 vs 4.5 pA); is this physiologically meaningful? Although the authors say the decrease in mini freq is not significant in Fig. 3B after PTX, it does appear rather large, a 40% reduction (5 vs 3 Hz).*

We must be guided by statistics in drawing conclusions, but the reader can interpret our data as they wish. Minis measure quantal release and thus to appreciate how small change can, when combined over the many receptors present, influence cell physiology, one needs to compare spiking activity. We show in Fig 2 that such change is sufficient to increase the excitatory synaptic drive provided by the A18a neuron. The seemingly larger reduction in mini frequency is intriguing and may reflect additional change, but without further experiments we cannot draw firm conclusions.

1. *The clever vibration assay is a good one to induce the activation of mechanosensory neurons, but the specificity of the changes induced by this is difficult to ascertain. One possibility would be to silence the output of the ch neurons (by expression to tetanus or botulinum toxin) and still put the larvae through the same vibration during the CP to see if the rescue is lost.*

We agree that further experiments are required to fully understand underlying mechanism(s). However, we will not be able to complete such follow-on expts in a timely manner and thus, these must wait and form the basis of future studies.

Minor points

1. *Typos - there are numerous areas where it seems a comma is used inappropriately (e.g. lines 28, 69, 77, 104, 348, 365, etc). Suggest line editing the final "version of record".*

Checked and corrected.

1. It would be of benefit to show the genotypes of the larvae in the various experimental manipulations in the relevant figure legends. This reviewer could not follow exactly how each experiment was done as it was not always clear which driver was being used to express which transgene in what genetic background.

Done

Reviewer #3 (Recommendations For The Authors):

• Please provide sample videos of electroshock-induced seizures (e.g. Fig 1B). Is it clear that the period of immobility after electroshock is a seizure (perhaps defined as hyperactivity originating from the brain)? I acknowledge the Baines group is quite skilled in this technique and perhaps there is a straightforward answer or citation to include.

We refer the reader to Marley and Baines 2011 which contains videos of seizure activity (first paragraph of Results).

• Seizures are generated in the brain and travel to the periphery. Do the authors think it is possible that the peripheral manipulations in this manuscript might be controlling the behavioral readout of seizures without affecting hypersynchronous activity in the brain?

We include the following statement (in methods) to provide our best understanding for how peripheral electroshock induces seizure..... ‘Strong peripheral stimulation likely causes excessive and synchronous synaptic excitation within the CNS resulting in seizure. However, the precise mechanism of this effect remains to be determined.’ Moreover, we feel it unlikely that manipulation of Ch neurons, by vibration, would suppress the effects we observe via peripheral mechanisms. Indeed, the Ch manipulation is limited to the embryonic CP, whilst our seizure assays are recorded many days later at L3.

• How might enhancement of inhibition lead to worsened seizures? Is the enhancement of ch-related inhibition selectively affecting inhibitory circuits, thereby leading to a net increase in excitation?

This is a difficult point to respond to at present. Enhanced inhibition per se might similarly disturb the encoding of an appropriate homeostatic setpoint(s) thus leaving a network open to being destabilized by a strong stimulus. Indeed, we have previously shown that increased inhibition during the CP results in the same effect (seizure) as increasing excitation (Giachello and Baines, 2015). Thus, presuming activation of Ch neurons during the CP translates to increased inhibition, then worsened seizure behaviour is a predictable effect. How this is achieved remains unknown and we prefer not to speculate here.